

# Gluon Multiplicity Moments and Particle-Number Fluctuations at Order $g^4$

A formalism, feasibility and publication analysis of the programme announced in  
*Soft Gluon Wave Function and Evolution Operator in the CGC at Next-to-Leading Order*

[arXiv:2607.18373](#) · R. Radhakrishnan

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**Three structural results.** (1) Because  $\Omega$  is written fully normal-ordered, its C, D and E structures annihilate the soft vacuum, so  $\Omega|0\rangle$  is exactly a truncated displaced squeezed state through  $O(g^2)$  — and every moment follows from a Fredholm determinant. (2) Because  $\langle aa \rangle_c$  is  $O(g^2)$  while  $\langle a^\dagger a \rangle_c$  is  $O(g^4)$ , the leading non-Poissonian effect is displacement-squeezing interference, and  $C_2$  is complete at  $O(g^4)$  with the existing operator. (3) Bose enhancement peaks near-side; pair emission from the squeezing kernel  $B_2$  peaks away-side — the only clean discriminator against classical colour-charge fluctuations, which are degenerate in both  $g$  and  $1/N_{\text{dof}}$ .

**Two limits stated honestly.**  $C_3$  and  $C_4$  are *not* complete — the three-gluon component of  $\Omega|0\rangle$  first appears at  $O(g^3)$  and contributes at the same order. And the classical and quantum contributions cannot be separated by power counting.

**Verdict.** One PRD paper, ~70% confidence, conditional on a two-week go/no-go test. Not JHEP as scoped. More than a note.

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# Scope, notation, and honest caveats

This directory contains a research analysis of the program announced at the end of the Introduction of

*R. Radhakrishnan, Soft Gluon Wave Function and Evolution Operator in the CGC at Next-to-Leading Order, arXiv:2607.18373 [hep-ph] (2026),*

namely: **moments of the gluon multiplicity operator and particle-number fluctuations at order  $g^4$ .**

## Status — reconciled against the paper (v2, 57 pp.)

Parts I–IX were first drafted without the paper (session egress blocked arxiv.org and all mirrors). The paper has since been read. **The structural conclusions survived; the coefficient dictionary did not.** This section records both, because the corrections matter.

### What was wrong

I guessed that the five labels  $A_{\dots}E$  ran over the five Fock structures available to an  $O(g^2)$  normal-ordered operator ( $a^\dagger, a, a^\dagger a, a^\dagger a^\dagger, aa$ ). **They do not.** The paper's Eq. (4.1) organizes by *operator length*, not by order in  $g$ , and one of the coefficients sits at  $O(g)$  with three operators — a possibility I did not anticipate:

| Paper (Eq. 4.1)       | Fock structure                         | Order            | Content                                                                           |
|-----------------------|----------------------------------------|------------------|-----------------------------------------------------------------------------------|
| $N$ (blackboard $N$ ) | $1$                                    | $N - 1 = O(g^2)$ | normalization / virtual correction                                                |
| $A_1, A_2$            | $a, a^\dagger$                         | $O(g)$           | soft emission/absorption off the valence charge, $\propto p$                      |
| $B_1, B_2, B_3$       | $aa, a^\dagger a^\dagger, a^\dagger a$ | $O(g^2)$         | pair emission, pair absorption, rescattering                                      |
| $C_1, C_2$            | $a^\dagger a^\dagger a, a^\dagger aa$  | $O(g)$           | <b>soft→soft splitting from the three-gluon vertex; carries no <math>p</math></b> |
| $D_{1,2,3}$           | four operators                         | $O(g^2)$         |                                                                                   |
| $E_{1,2,3}$           | six operators                          | $O(g^2)$         | built from $C \cdot C$                                                            |

The  $O(g)$  trilinear  $C$  is the substantive surprise. It means  $\Omega$  **is not a Gaussian (quadratic-exponent) unitary at any order** — my §1.5 claim, as a statement about  $\Omega$ , is false.

### What survived, and why

The Gaussian machinery is nevertheless *correct for the observable*, for a reason that is now verifiable rather than assumed.  $\Omega$  in Eq. (4.1) is written **fully normal-ordered**, so every  $C, D, E$  structure carries at least one annihilation operator on the right and annihilates the soft vacuum. Acting on  $|0\rangle$ , only three terms survive:

$$\Omega|0\rangle = N|0\rangle + \int A_2 a^\dagger|0\rangle + \iint B_2 a^\dagger a^\dagger|0\rangle \quad (\text{exact through } O(g^2))$$

which is precisely a truncated **displaced squeezed state**. So the identifications used throughout Parts II–IV hold, with this dictionary:

| this document              | paper (Eq. 4.1, 5.1–5.3) | order                         |
|----------------------------|--------------------------|-------------------------------|
| $\alpha$ (displacement) →  | $A_2$ [Eq. (5.2)]        | $O(g), \propto p$             |
| $m$ (pair/anomalous)→      | $B_2$ [Eq. (5.3)]        | $O(g^2), \propto pp \oplus p$ |
| $n$ (occupation) →         | $B_2^\dagger B_2$        | $O(g^4)$                      |
| $Z$ (normalization) →      | $N$ [Eq. (5.1)]          | $1 + O(g^2)$                  |
| my "A" →                   | $A_2$ my "D" → $B_2$     |                               |
| my "B" →                   | $A_1$ my "E" → $B_1$     |                               |
| my "C" ( $a^\dagger a$ ) → | $B_3$                    |                               |

Eq. (5.2) is  $A_2{}^b{}_j(p) = -\sqrt{2} g p^j p^b(-p)/(\sqrt{p^+ p^2})$  — the Weizsäcker–Williams field, as assumed. Eq. (5.3) for  $B_2$  contains exactly the two pieces predicted in §1.2: a  $pp$  two-source term and an  $f^{\{acb\}}p$  term from the three-gluon vertex, plus an instantaneous  $H_{gg-inst}$  contribution.

**Consequences for the research program:**

- The headline formula is unchanged, now with verified labels:  $C_2 = 2\text{Re} \int A_2^* B_2 A_2^* + \int |B_2|^2$ , complete at  $O(g^4)$ .
- $C_3$  incompleteness is **confirmed**:  $\Omega|0\rangle$  has no three-gluon component at  $O(g^2)$ , so the three-gluon amplitude first appears at  $O(g^3)$  and is not in this operator.
- **The wave-function multiplicity program needs only Eqs. (5.1)–(5.3)** — three of the paper's coefficients.  $C, D, E$  are irrelevant to  $\Omega|0\rangle$ . They become essential the moment the target Wilson lines act ( $a \rightarrow U a$ ), because then the annihilation operators no longer kill the state. That is Project E in Part VIII, and it is the natural home for the rest of the operator.

What is still approximate here

Numerical prefactors, phases, and the light-cone normalization of  $a, a^\dagger$  are not reproduced in Parts I–IV. Formulas are correct in structure, scaling and parametric order; assembling them into publishable expressions still requires one pass against Eqs. (5.1)–(5.3).

Caveat 2 — what I am *not* doing

I am not summarizing the paper. Parts I–IV build the observable-side formalism the paper does not contain; Parts V–IX are a research assessment. Part VIII is deliberately unflattering where the physics warrants it.

Conventions used throughout

| Symbol                                                                                  | Meaning                                                                                                |
|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| $\rho^a(x)$                                                                             | valence color charge density operator, transverse position $x$ , adjoint index $a$                     |
| $[\rho^a(x), \rho^b(y)] = i g f^{abc} \delta^2(x-y) \rho^c(x)$                          | valence charge algebra ( $su(N_c)$ current algebra)                                                    |
| $a^{\{a\}}_{i\}(k)$                                                                     | soft gluon creation operator: transverse momentum $k$ , color $a$ , transverse polarization $i = 1, 2$ |
| $[a^a_{i\}(k), a^{\{b\}}_{j\}(q)] = (2\pi)^2 \delta^2(k-q) \delta^{\{ab\}} \delta_{ij}$ | soft Fock algebra (light-cone normalization absorbed)                                                  |
| $b^a_{i\}(k) = i g (k_i / k^2) \rho^a(k)$                                               | Weizsäcker–Williams field (momentum space), $O(g)$                                                     |
| $N = \int_k a^{\{a\}}_{i\}(k) a^a_{i\}(k)$                                              | soft gluon number operator                                                                             |
| $\langle \cdot \rangle_\rho$                                                            | expectation in the soft Fock space at <b>fixed</b> valence configuration                               |
| $\langle \langle \cdot \rangle \rangle$                                                 | additionally averaged over the CGC weight functional $W[\rho]$                                         |
| $S_\perp$                                                                               | transverse area; $Q_s$ saturation scale; $\mu^2$ MV color charge density                               |
| dilute counting                                                                         | $\rho = O(1)$ , $g$ explicit                                                                           |
| dense counting                                                                          | $g \rho = O(1)$ , i.e. $Q_s \gg \Lambda_{QCD}$                                                         |

Symbols introduced later, listed here so they are not mistaken for the paper's own notation:  $w, w^-$  = Nambu-space displacement vector ( $\alpha, \alpha^*$ ) and its conjugate;  $K$  = Nambu covariance;  $\Gamma(k, q)$  = connected two-gluon correlation;  $\theta$  = the  $\rho$ -stripped splitting kernel;  $\Xi$  = the anti-hermitian generator in  $\Omega = \exp(\Xi)$ ;  $Z[\rho]$  = the normalization of  $\Omega$ .

Powers of  $g$  are quoted in **dilute counting** unless stated. Where the distinction matters — and it matters a great deal for the publication case — both countings are given.

File map

| File               | Content                                                                               |
|--------------------|---------------------------------------------------------------------------------------|
| 01-formalism.md    | Part I — structure of $\Omega$ , Bogoliubov reduction, what is needed for observables |
| 02-moments.md      | Part II — $\langle N \rangle \dots \langle N^4 \rangle$                               |
| 03-cumulants.md    | Part III — $K_2, K_3, K_4$ , factorial cumulants, what is genuinely new               |
| 04-correlations.md | Part IV — $n$ -gluon distributions, generating function                               |
| 05-models.md       | Part V — model calculations and their integrals                                       |
| 06-numeric.md      | Part VI — proposed plots                                                              |
| 07-literature.md   | Part VII — literature survey, what is already done                                    |

| File                  | Content                                             |
|-----------------------|-----------------------------------------------------|
| 08-publication.md     | Part VIII — ranked, critical publication assessment |
| 09-roadmap.md         | Part IX — concrete research plan                    |
| 10-jhep-assessment.md | Addendum — will the paper itself make it into JHEP? |

# Part I — The formalism you actually need to compute observables

The paper builds  $\Omega$ . It does not build the machinery that turns  $\Omega$  into a multiplicity distribution. That machinery is the subject of this file, and it is the real content of the proposed follow-up: **the hard part of the multiplicity project is not the  $O(g^4)$  diagrams — those are already encoded in  $\Omega$  — it is knowing which structural reduction makes the Fock-space algebra finite**. There is one, and it is the reason this project is tractable at all.

## 1.1 The state

Born–Oppenheimer separation splits the hadron's light-cone wave function into a fast valence sector ( $k^+ > \Lambda$ ) and a soft sector ( $k^+ < \Lambda$ ). The soft modes are created on top of the valence state by a unitary dressing operator:

$$|\Psi\rangle = \Omega[p; a, a^\dagger] |v\rangle \otimes |0\rangle_{\text{soft}}$$

Two facts drive everything:

1.  $\Omega$  is unitary in the soft Fock space —  $\Omega^\dagger \Omega = 1$ . This is what fixes the coefficient relations the paper uses.
2.  $\Omega$ 's coefficients are operators, not numbers. They are functionals of  $p^\perp a(x)$ , which does not commute with itself. So the coefficients must be kept in order throughout.

## 1.2 The normal-ordered expansion (as actually written in the paper)

Eq. (4.1) of arXiv:2607.18373 organizes  $\Omega$  by **operator length**, not by order in  $g$ :

$$\begin{aligned} \Omega = & N \\ & + \int_p [A_1 a + A_2 a^\dagger] \\ & + \int_{\{p,q\}} [B_1 aa + B_2 a^\dagger a^\dagger + B_3 a^\dagger a] \\ & + \int_{\{p,q,r\}} [C_1 a^\dagger a^\dagger a + C_2 a^\dagger aa] \\ & + \int_{\{p,q,r,s\}} [D_1 a^\dagger a^\dagger a^\dagger a, D_2 a^\dagger a^\dagger aa, D_3 a^\dagger aaaa] \\ & + \int_{\{p,\dots,v\}} [E_{1,2,3} \text{ six-operator structures}] \end{aligned}$$

(colour, polarization and momentum labels suppressed; the paper writes these in blackboard bold  $N, A, B, C, D, E$ .)

**Order counting** — the paper's assignment, which is *not* what a naive Fock-structure argument would give:

| Coefficient     | Fock structure                         | Order            | Content                                                                                       |
|-----------------|----------------------------------------|------------------|-----------------------------------------------------------------------------------------------|
| $N$             | 1                                      | $N - 1 = O(g^2)$ | normalization = the virtual correction                                                        |
| $A_1, A_2$      | $a, a^\dagger$                         | $O(g)$           | soft emission off the valence charge; $A_2 \propto p$ , Eq. (5.2) = the WW field              |
| $B_1, B_2, B_3$ | $aa, a^\dagger a^\dagger, a^\dagger a$ | $O(g^2)$         | pair absorption, <b>pair emission (squeezing)</b> , rescattering                              |
| $C_1, C_2$      | $a^\dagger a^\dagger a, a^\dagger aa$  | $O(g)$           | <b>soft→soft splitting from the three-gluon vertex; carries no <math>p</math></b> , Eq. (5.5) |
| $D_{1,2,3}$     | four operators                         | $O(g^2)$         | partly expressible via $A$ and $C$ , Eqs. (5.6)–(5.7)                                         |
| $E_{1,2,3}$     | six operators                          | $O(g^2)$         | built from $C \cdot C$ , Eqs. (5.8)–(5.9)                                                     |

**The  $O(g)$  trilinear  $C$  is the structurally important feature.** It exists because the three-gluon vertex connects one soft gluon to two soft gluons with no valence insertion at all — a transition that costs one power of  $g$  and zero powers of  $p$ . Its presence means  $\Omega$  is **not a Gaussian unitary at any order**, and it is what makes the  $O(g)$  diagonalization in the paper's §6 nontrivial: the coherent operator alone does not diagonalize  $H_1$ ;  $C$  is needed too.

**The two-piece split of  $B_2$  is the physics of the fluctuation program.** Eq. (5.3) contains a  $pp$  term — "two independent emissions correlated only through the source" — and an  $f^{\{acb\}}p$  term from the three-gluon vertex, the genuine  $1 \rightarrow 2$  splitting off a single charge, plus an instantaneous piece from  $H_{gg-inst}$ . Only the second is irreducibly quantum. **Any claim of new physics in the fluctuations should be traced to which piece is responsible.**

**Completeness of the basis.** The structures omitted from (4.1) are the ones forbidden by  $k^+$  conservation ( $a\bar{t}a\bar{t}a\bar{t}$ ,  $aaa$ ,  $a\bar{t}a\bar{t}a\bar{t}a\bar{t}$ , ...) or first arising at  $O(g^3)$  ( $a\bar{t}a\bar{t}a\bar{t}a\bar{t}a\bar{t}$ , ...). A genuinely irreducible  $2\rightarrow 3$  transition would need a five-operator coefficient at  $O(g)$ ; it does not exist, because  $2\rightarrow 3$  is already generated by  $C_1$  acting with one spectator gluon. The basis is complete at  $O(g^2)$ .

### 1.3 Unitarity constraints

$\Omega^\dagger\Omega = 1$  at each order in  $g$  relates the coefficients. The paper's version is Eqs. (4.3)–(4.5):  $\Omega^\dagger{}^{(1)} = -\Omega^{(1)}$  at  $O(g)$ , and  $\Omega^{(2)} + \Omega^\dagger{}^{(2)} = -\Omega^\dagger{}^{(1)}\Omega^{(1)}$  at  $O(g^2)$ . Concretely:

- $A_1 = -A_2^*$  and  $B_1 = -B_2^*$  (emission/absorption pairs),
- $B_3$  and  $D_3$ ,  $E_3$  are *fixed by the constraint equation* rather than by matching — the paper states this explicitly below Eq. (5.12),
- $N$  is fixed by  $\langle\Psi|\Psi\rangle = 1$ ; Eq. (5.1) gives  $N = 1 - g^2\int \rho^b(-p)\rho^b(p)/(p^+p^2)$ . **The normalization *is* the virtual correction**, and it must be kept — it is exactly what makes  $\langle N\rangle$  finite mode-by-mode.
- The paper also writes  $\Omega = \exp(iG)$  with  $G$  Hermitian by construction, Eq. (5.11)–(5.12).

**Practical consequence for the observable:** although  $\Omega$  has six coefficient families, the soft-vacuum sector sees only three of them. See §1.4.

### 1.4 Dictionary — verified against Eq. (4.1) and Eqs. (5.1)–(5.3)

| this document |                        | paper                                                | order                                   |
|---------------|------------------------|------------------------------------------------------|-----------------------------------------|
| A             | ( $a\bar{t}$ )         | -> $A_2$ [Eq. (5.2)]                                 | $O(g)$ , $\propto p^-$ (WW field)       |
| B             | ( $a$ )                | -> $A_1$                                             | $O(g)$                                  |
| C             | ( $a\bar{t}a$ )        | -> $B_3$ [Eq. (5.4)]                                 | $O(g^2)$                                |
| D             | ( $a\bar{t}a\bar{t}$ ) | -> $B_2$ [Eq. (5.3)]                                 | $O(g^2)$ , $\propto p\rho^+ \oplus p^-$ |
| E             | ( $aa$ )               | -> $B_1$                                             | $O(g^2)$                                |
| Z             | (normaliz.)            | -> $N$ [Eq. (5.1)]                                   | $1 + O(g^2)$                            |
| –             | (no analogue)          | -> $C_1, C_2$ ( $a\bar{t}a\bar{t}a$ , $a\bar{t}aa$ ) | $O(g)$ ← I did not anticipate this      |
| –             |                        | -> $D_{1,2,3}$ ; $E_{1,2,3}$                         | $O(g^2)$                                |

Read Parts II–IV with  $A \rightarrow A_2$  and  $D \rightarrow B_2$  throughout. Every formula below is unaffected.

### 1.5 The key structural reduction

The claim in the first draft — “ $\Omega$  is a Gaussian unitary at fixed  $\rho$ ” — is **false**.  $\Omega$  contains the  $O(g)$  trilinear  $C$  and the  $O(g^2)$  quartic and sextic structures  $D$ ,  $E$ ; it is not the exponential of a quadratic form.

The conclusion nevertheless holds for the observable, for a sharper reason.  $\Omega$  **is written fully normal-ordered**, so every  $C$ ,  $D$ ,  $E$  structure carries at least one annihilation operator on the right and therefore annihilates the soft vacuum. Acting on  $|\theta\rangle$ :

$$\Omega|\theta\rangle = N|\theta\rangle + \int A_2 a^\dagger|\theta\rangle + \iint B_2 a\bar{t}a^\dagger|\theta\rangle$$

exact through  $O(g^2)$

which has 0-, 1- and 2-gluon components only — i.e. it agrees with a **displaced squeezed vacuum** through the order retained. So the Gaussian machinery below is not an approximation to  $\Omega$ ; it is an exact parameterization of  $\Omega|\theta\rangle$  at this order, and it fails at  $O(g^3)$ , where a three-gluon component appears.

Two consequences worth stating plainly:

- The wave-function multiplicity program needs only Eqs. (5.1), (5.2), (5.3).** The paper's  $C$ ,  $D$ ,  $E$  play no role in  $\Omega|\theta\rangle$ .
- They become essential the moment the target acts.** After eikonal scattering  $a \rightarrow U a$ , the annihilation operators no longer kill the state, and  $C$ ,  $D$ ,  $E$  all contribute to produced- gluon observables. That is Project E of Part VIII, and it is the natural consumer of the rest of the paper's results.

With that established: the soft state is Gaussian through  $O(g^2)$ , so

$$\Omega^\dagger a^\dagger a_i(k) \Omega = \int_{-q} [ u(k,q) a(q) + v(k,q) a^\dagger(q) ] + \alpha(k)$$



an *inhomogeneous Bogoliubov transformation*, with (schematically, suppressing colour and polarization indices which ride along)

$$\begin{array}{ll} \alpha = A_2 + 0(g^3) & [\text{displacement}, 0(g)] \\ v = B_2 + 0(g^4) & [\text{pair amplitude}, 0(g^2)] \\ u = 1 + 0(g^2) & [0(1) + 0(g^2)] \end{array}$$

and unitarity becomes the standard Bogoliubov relations  $u u^\dagger - v v^\dagger = 1, u v^\dagger = v u^\dagger$ .

**Everything about  $N$  follows from three objects**, evaluated at fixed  $\rho$ :

$$\begin{array}{llll} \alpha_k & = \langle \Psi | a_k | \Psi \rangle & \text{displacement} & 0(g) \\ n_{\{kl\}} & = \langle \Psi | a_k^\dagger a_l | \Psi \rangle - \alpha_k^* \alpha_l & \text{connected occupation} & 0(g^4) = (v v^\dagger)_{\{kl\}} \\ m_{\{kl\}} & = \langle \Psi | a_k a_l | \Psi \rangle - \alpha_k \alpha_l & \text{anomalous / pair} & 0(g^2) = (u v^\dagger)_{\{kl\}} \end{array}$$

(indices  $k$  collapse momentum + colour + polarization.)

**The order assignment  $m = 0(g^2)$  while  $n = 0(g^4)$  is the pivot of this entire research program.** The anomalous correlator is *one* power of the pair amplitude; the occupation correction is *two*. Consequently the leading non-Poissonian effect is an **interference between the classical Weizsäcker–Williams displacement and the quantum pair-emission amplitude**, of order  $\alpha^* \alpha m \sim g^2 \cdot g^2 = g^4$  — precisely the order the paper's Introduction promises. It is *not* the  $|B_2|^2$  "two real gluons" term, which is  $g^8$ -suppressed relative to  $\langle N \rangle^2$ . Anyone attacking this problem by squaring the two-gluon emission amplitude is computing a subleading effect.

## 1.6 How $\Omega$ acts on Fock states

For completeness, since the moments can also be organized as a Fock-space sum:

$$\Omega|0\rangle = N [ |0\rangle + \int A_2 a^\dagger |0\rangle + \frac{1}{2} \int \int (A_2 A_2 + B_2) a^\dagger a^\dagger |0\rangle + \dots ]$$

so the  $n$ -gluon component of the dressed state is

$$\Psi_n(k_1 \dots k_n) = N \cdot \text{Sym} [ \prod A_2(k_i) + \sum_{\{\text{pairs}\}} B_2(k_i, k_j) \prod_{\{\text{rest}\}} A_2 + \dots ]$$

i.e. a **coherent (product) part plus pairwise-correlated insertions**. This is the Fock-space statement of "displaced squeezed state" and is the form in which the CGC literature usually writes things. It is *much* worse for computing high moments than the Bogoliubov route — the combinatorics of **Sym** explodes — which is why §1.5 is the recommended organizing principle. Use the Fock form only for cross-checks at  $n \leq 3$ .

## 1.7 Where the valence charge operators bite

Two distinct roles, often conflated:

- (a)  **$\rho$  as a source (classical).**  $A_2 \propto b \propto g\rho/k$ . This is what survives when  $[\rho, \rho] \rightarrow 0$  and gives the classical CGC picture.
- (b)  **$\rho$  as a non-commuting operator.**  $[\rho^a(x), \rho^b(y)] = ig f^{abc} \delta^2(x-y) \rho^c(x)$ . Because  $f^{abc}$  carries an explicit  $g$ , a single commutator costs  $g$  and converts  $\rho\rho \rightarrow \rho$ . So:

- In  $\langle N \rangle$  at  $0(g^4)$ , reordering products produces  $g \cdot \langle \rho \rho \rho \rangle \rightarrow \langle \rho \rho \rangle$  terms that are the *same order* as genuine two-loop-like contributions. **They cannot be dropped.**
- This is where the "non-commuting valence colour charge operators" emphasized in the abstract actually do work. It is also the single most likely source of an algebra error, and the most likely target of a referee's technical objection. (The paper's own Eq. (4.28)–(4.29) "decomposition procedure" is exactly this step, done correctly.)

**Ordering prescription.** You must adopt and state one — e.g. all  $\rho$ 's to the right of all  $a^\dagger$ 's, symmetrized in the valence indices — and show that observables are independent of it (they must be, since  $\Omega$  is a fixed operator; the ordering only redistributes content between coefficients). A short appendix demonstrating ordering-independence of  $\langle N \rangle$  at  $0(g^4)$  would pre-empt an obvious referee complaint.

## 1.8 How expectation values are computed — the two-stage structure

$$\langle f(N) \rangle = \int d\rho \underbrace{W_Y[\rho]}_{\text{stage 2}} \underbrace{\langle f(N) \rangle_\rho}_{\text{stage 1}}$$

- **Stage 1 (Fock).** At fixed  $\rho$ , the state is Gaussian through  $0(g^2) \Rightarrow$  Wick's theorem in  $(\alpha, n, m)$  gives every moment in closed form. This stage is *exactly solvable*.
- **Stage 2 (colour).**  $W_Y[\rho]$  is the CGC weight: MV (Gaussian) at the initial condition, JIMWLK-evolved thereafter. This stage produces  $\langle \rho^{2p} \rangle$  correlators.

The correct way to combine them is the **law of total cumulance**, which cleanly separates "quantum Fock-space fluctuation" from "classical colour-charge fluctuation":

$$\begin{aligned} \kappa_1 &= E_\rho[\mu(\rho)] \\ \kappa_2 &= E_\rho[\kappa_2^{\text{Fock}}(\rho)] + \text{Var}_\rho[\mu(\rho)] \\ \kappa_3 &= E_\rho[\kappa_3^{\text{Fock}}(\rho)] + 3 \text{Cov}_\rho[\mu(\rho), \kappa_2^{\text{Fock}}(\rho)] + \kappa_3^\rho[\mu(\rho)] \\ \kappa_4 &= E_\rho[\kappa_4^{\text{Fock}}] + 4\text{Cov}_\rho[\mu, \kappa_3^{\text{Fock}}] + 3\text{Var}_\rho[\kappa_2^{\text{Fock}}] \\ &\quad + 6 \kappa_3^\rho\text{-type mixed terms} + \kappa_4^\rho[\mu(\rho)] \end{aligned}$$

with  $\mu(\rho) \equiv \langle N \rangle_\rho$ .

This decomposition is, in my judgement, **the single most publishable organizational idea in the whole program**, because it makes the following statement provable rather than rhetorical:

Every term containing  $\kappa_\rho^\rho[\mu]$  is classical CGC colour-charge fluctuation — known physics, parametrically enhanced by the number of sources. Every term containing  $\kappa_\rho^{\text{Fock}}$  is genuinely quantum, generated by the  $0(g^2)$  pair-emission coefficient  $B_2$  of  $\Omega$ , and is not enhanced by the source density. The two are distinguishable by their  $N_c, S_\perp$ , and  $k_\perp$  dependence.

## 1.9 Which correlators of $\rho$ you will need

| Observable                      | Highest $\rho$ correlator             | MV (Gaussian) status                                                                   |
|---------------------------------|---------------------------------------|----------------------------------------------------------------------------------------|
| $\langle N \rangle$ at $0(g^2)$ | $\langle \rho \rho \rangle$           | $= g^2 \mu^2 \delta^{ab} \delta^2(x-y)$ , trivial                                      |
| $\langle N \rangle$ at $0(g^3)$ | $\langle \rho \rho \rho \rangle$      | <b>vanishes</b> in MV (C-even weight); nonzero only with odderon / $d^{abc}$ odd terms |
| $\langle N \rangle$ at $0(g^4)$ | $\langle \rho \rho \rho \rho \rangle$ | Wick $\rightarrow$ 3 pairings                                                          |
| $\langle N^2 \rangle$           | $\langle \rho^4 \rangle$              | 3 pairings                                                                             |
| $\langle N^3 \rangle$           | $\langle \rho^6 \rangle$              | 15 pairings                                                                            |
| $\langle N^4 \rangle$           | $\langle \rho^8 \rangle$              | 105 pairings                                                                           |

The  $(2p-1)!!$  growth looks alarming but is not the real cost: in cumulants all disconnected pairings cancel by construction, and the surviving connected topologies number  $0(p!)$  at worst. The genuine cost is that each connected topology carries a distinct transverse-momentum convolution (see Part V).

**The  $\langle N \rangle$  at  $0(g^3)$  entry is worth pausing on.** It is the first place a C-odd (odderon) correlator enters the gluon multiplicity. In MV it is zero; in a model with a nonzero  $\langle \rho^a \rho^b \rho^c \rangle$   $d^{abc}$  it is not. That is a small, self-contained, genuinely novel calculation — see Part VIII, Project D.

## Summary of Part I

- $\Omega$  is *not* Gaussian — the  $0(g)$  trilinear  $C$  and the  $0(g^2)$   $D, E$  see to that. But it is written fully normal-ordered, so  $\Omega|0\rangle$  is exactly a truncated displaced squeezed state through  $0(g^2)$ , parameterized by  $N, A_2, B_2$  alone.
- At fixed classical  $\rho$ , all moments follow from  $(\alpha, n, m)$  by Wick's theorem. Exactly solvable.
- $m = 0(g^2)$  but  $n = 0(g^4)$ . The leading non-Poissonian effect is the **displacement-squeezing interference**  $\alpha^* \alpha^* m \sim 0(g^4)$ , not  $|B_2|^2$ .
- $\rho$  non-commutativity contributes at the *same* order as the NLO kernels and must be retained; an ordering-independence check is mandatory.

5. The law of total cumulance separates quantum (Fock) from classical (colour) fluctuations. Use it as the paper's organizing principle.

# Part II — Moments of the gluon multiplicity operator

## 2.0 Strategy

Do **not** compute  $\langle N \rangle$ ,  $\langle N^2 \rangle$ ,  $\langle N^3 \rangle$ ,  $\langle N^4 \rangle$  by brute-force Wick contraction of  $\Omega^\dagger N^p \Omega$ . The combinatorics grow like  $(2p)!$  and the  $N^p$  products generate a forest of contact terms from  $[a, a^\dagger]$  that all cancel in the end.

Instead:

1. Compute the **factorial moments**  $F_p = \langle :N^p: \rangle$  — normal-ordered, so no contact terms.
2. Get them from the **generating function**, which at fixed  $p$  is a closed-form Fredholm determinant (§2.2). This is exact, not perturbative.
3. Convert to ordinary moments by Stirling numbers (§2.5).

This reduces "compute the fourth moment" from a diagrammatic nightmare to evaluating  $\text{Tr}(K^4)$  and  $w^{-1}K^3w$  for two known kernels.

## 2.1 The three objects

Everything reduces to (fixed  $p$ ; index  $k$  collects  $\{k_\perp, a, i\}$ ;  $\int_k \equiv \int d^2k/(2\pi)^2$  with colour/polarization sums):

$$\begin{aligned} \alpha_k &= \langle \Psi | a_k | \Psi \rangle &= A_{2,k} &+ O(g^3) &[O(g)] \\ m_{\{kl\}} &= \langle \Psi | a_k a_l | \Psi \rangle - \alpha_k \alpha_l &= B_{2,\{kl\}} &+ O(g^4) &[O(g^2)] \\ n_{\{kl\}} &= \langle \Psi | a^\dagger_k a_l | \Psi \rangle - \alpha^*_k \alpha_l &= (B_2^\dagger B_2)_{\{kl\}} &+ O(g^6) &[O(g^4)] \end{aligned}$$

$A_2$  and  $B_2$  are the paper's coefficients, Eqs. (5.2) and (5.3); see `00-scope §Status` for the dictionary.  $A_{2^b,1}(p) = -\sqrt{2} g p^b p(-p)/(\sqrt{p^+ p^2})$  is the Weizsäcker–Williams field.

Derivation: these are read straight off  $\Omega|0\rangle = N|0\rangle + \int A_2 a^\dagger |0\rangle + \iint B_2 a^\dagger a^\dagger |0\rangle$  (Part I §1.5), which is exact through  $O(g^2)$  because the fully normal-ordered  $C, D, E$  structures all annihilate the soft vacuum. Equivalently, in Bogoliubov language  $\Omega^\dagger a_k \Omega = \int_q [u_{\{kq\}} a_q + v_{\{kq\}} a^\dagger_q] + \alpha_k$  with  $\alpha = A_2 + O(g^3)$ ,  $v = B_2 + O(g^4)$ ,  $u = 1 + O(g^2)$ ; unitarity ( $A_1 = -A_2^*$ ,  $B_1 = -B_2^*$ , Eqs. 4.4–4.5) guarantees  $uu^\dagger - vv^\dagger = 1$ .

**Order hierarchy — the central fact.**

$$\begin{array}{ll} m = O(g^2) & \text{is ONE power of the pair amplitude} \\ n = O(g^4) & \text{is TWO powers} \end{array}$$

Consequently the leading fluctuation effect is the **interference of the classical Weizsäcker–Williams displacement with the quantum pair-emission amplitude**, not the modulus squared of the two-gluon amplitude. This determines which diagrams matter and is the reason the promised  $O(g^4)$  calculation is feasible with an  $O(g^2)$  operator.

## 2.2 Exact generating function at fixed $p$

Assemble the Nambu-space displacement and covariance

$$w = (\alpha, \alpha^*)^\top, \quad w^{-1} = (\alpha^*, \alpha), \quad K = \begin{bmatrix} [n, m] & [m^*, n^\top] \end{bmatrix}$$

Then for the displaced squeezed (Gaussian) soft state,

$$G(z) \equiv \langle z^N \rangle_p = \det[1 - (z-1)K]^{(-1/2)} \cdot \exp \left\{ \frac{(z-1)}{2} w^{-1} [1 - (z-1)K]^{(-1)} w \right\}$$

This is **exact for Gaussian**  $\Omega|0\rangle$ , i.e. exact at fixed classical  $p$  through  $O(g^2)$ , to all orders in the kernels. Checks:

| Limit              | $G(z)$                  | Distribution |
|--------------------|-------------------------|--------------|
| $K = 0$ (coherent) | $\exp[(z-1) \alpha ^2]$ | Poisson ✓    |

| Limit                         | $G(z)$                                   | Distribution     |
|-------------------------------|------------------------------------------|------------------|
| $\alpha = 0, m = 0$ (thermal) | $\det[1 + (1-z)n]^{-1}$                  | Bose-Einstein ✓  |
| single-mode squeezed vacuum   | $[\cosh^2 r - z^2 \sinh^2 r]^{-1/2}$     | even- $n$ only ✓ |
| displaced thermal             | $(1-t_n)^{-1} \exp[t \alpha ^2/(1-t_n)]$ | Laguerre ✓       |

All four limits reproduce the textbook quantum-optics results, so the formula is verified.

**Immediate corollary (factorial cumulants, all orders):** writing  $t = z-1$ ,

$$\begin{aligned} \log G(1+t) &= -\frac{1}{2} \text{Tr} \log(1 - tK) + \frac{(t/2)}{w} (1 - tK)^{-1} w \\ &= \frac{1}{2} \sum_{p \geq 1} t^p \text{Tr}(K^p)/p + \frac{1}{2} \sum_{p \geq 1} t^p w^{-1} K^{(p-1)} w \end{aligned}$$

hence

$$C_p = (p-1)!/2 \cdot \text{Tr}(K^p) + p!/2 \cdot w^{-1} K^{(p-1)} w$$

Since  $K = 0(g^2)$  and  $w = 0(g)$ ,  $C_p = 0(g^{\{2p\}})$  for every  $p$ . This is the clean statement of the order counting the paper's Introduction is pointing at.

## 2.3 First moment $\langle N \rangle$

$$\langle N \rangle_p = C_1 = \frac{1}{2} \text{Tr} K + \frac{1}{2} w^{-1} w = \text{Tr} n + |\alpha|^2$$

Expanded in  $g$ :

$$\begin{aligned} \langle N \rangle_p &= \int_k |A_2^{\{(1)\}}{}_k|^2 & 0(g^2) & \langle \rho\rho \rangle \\ &+ 2 \text{Re} \int_k A_2^{\{(1)\}*}{}_k A_2^{\{(2)\}}{}_k & 0(g^3) & \langle \rho\rho\rho \rangle \\ &+ \int_k |A_2^{\{(2)\}}{}_k|^2 + 2 \text{Re} \int_k A_2^{\{(1)\}*}{}_k A_2^{\{(3)\}}{}_k & & \\ &+ \int_{\{k,q\}} |B_{2,\{kq\}}|^2 & 0(g^4) & \langle \rho^4 \rangle \end{aligned}$$

**Reading of the three orders.**

- $0(g^2)$  — the classical Weizsäcker-Williams gluon number,  $\langle N \rangle_{LO} = \int d^2k/(2\pi)^2 (g^2/k^2) \langle \rho^a(k) \rho^a(-k) \rangle \cdot (N_c^2-1)$ -type. Log-divergent in the IR (regulated by  $Q_s$  once saturation is resummed) and in the UV (regulated by the  $k^+$  cutoff / matching to the hard sector). **Not new.**
- $0(g^3)$  — requires  $\langle \rho\rho\rho \rangle$ . **Vanishes identically for any C-even weight functional, in particular MV and Gaussian JIMWLK.** Nonzero only in the presence of an odderon-type odd correlator  $\langle \rho^a \rho^b \rho^c \rangle \propto d^{\{abc\}}$ . This is a small, clean, genuinely unexploited calculation.
- $0(g^4)$  — three physically distinct pieces:
  - $|A_2^{\{(2)\}}|^2$ : two-source emission squared — classical, enhanced by  $\rho^4$ ;
  - $2 \text{Re} A_2^{\{(1)\}*} A_2^{\{(3)\}}$ : virtual/self-energy correction interfering with the classical field — **this is where the  $\ln(1/x)$  rapidity divergence lives**, and it must exponentiate into JIMWLK acting on the LO result. Verifying that is a mandatory consistency check;
  - $\int |B_2|^2$ : real pair emission — the genuinely new quantum piece.

**Consistency requirement (do this first, before anything else).** The rapidity-divergent part of (2) plus the real emission (3) must combine into  $\partial_Y \langle N \rangle_{LO} = H_{JIMWLK} \triangleright \langle N \rangle_{LO}$ . If it does not, either the kernel extraction or the cutoff scheme is wrong. This check is cheap and it is the single most convincing technical validation you can put in a paper.

## 2.4 Higher factorial moments

From  $F_p$  in terms of  $C_p$  (exponential/Bell relations):

$$\begin{aligned}
F_1 &= C_1 \\
F_2 &= C_2 + C_1^2 \\
F_3 &= C_3 + 3 C_1 C_2 + C_1^3 \\
F_4 &= C_4 + 4 C_1 C_3 + 3 C_2^2 + 6 C_1^2 C_2 + C_1^4
\end{aligned}$$

with, from §2.2, at leading nontrivial order in  $g$ :

$$\begin{aligned}
C_2 &= \text{Tr}(n^2) + \text{Tr}(m m^\dagger) + 2 \alpha^\dagger n \alpha + 2 \text{Re}(\alpha^* m \alpha^*) \\
&\rightarrow \int_{\{k,q\}} |B_2, \{kq\}|^2 + 2 \text{Re} \int_{\{k,q\}} A_2^* \cdot k B_2, \{kq\} A_2 \cdot q \quad [0(g^4), \text{complete}] \\
C_3 &= \text{Tr}(K^3) + 3 w^- K^2 w \quad [0(g^6), \text{INCOMPLETE} - §2.6] \\
C_4 &= (3!/2) \text{Tr}(K^4) + 12 w^- K^3 w \quad [0(g^8), \text{INCOMPLETE} - §2.6]
\end{aligned}$$

Written out, the complete  $0(g^4)$  result is

$$\begin{aligned}
C_2 &= 2 \text{Re} \int_{\{k,q\}} A_2^{\{a^*\}} \cdot i(k) B_2^{\{ab\}} \cdot \{ij\}(k,q) A_2^{\{b^*\}} \cdot j(q) \\
&\quad + \int_{\{k,q\}} |B_2^{\{ab\}} \cdot \{ij\}(k,q)|^2 + 0(g^6)
\end{aligned}$$

This is the whole  $0(g^4)$  fluctuation calculation, at fixed  $p$ , in one line. Everything after this is colour averaging and numerics.

**Which colour correlators appear.** With  $A_2 \propto g p$  and  $B_2 \propto g^2 p p \oplus g^2 p$ :

| Term                                                                 | $p$ content                                               | MV status                        |
|----------------------------------------------------------------------|-----------------------------------------------------------|----------------------------------|
| $\text{Re } A_2^* B_2 A_2^*$ with $B_2 \propto p p$                  | $\langle p p p p \rangle$                                 | 3 Wick pairings                  |
| $\text{Re } A_2^* B_2 A_2^*$ with $B_2 \propto p(g \rightarrow g g)$ | $\langle p p p \rangle$                                   | <b>vanishes</b> in MV            |
| $\int  B_2 ^2$ with $B_2 \propto p p$                                | $\langle p p p p \rangle$                                 | 3 pairings                       |
| $\int  B_2 ^2$ with $B_2 \propto p$                                  | $\langle p p \rangle$                                     | trivial                          |
| $p$ -commutator reorderings                                          | $g \langle p p p \rangle \rightarrow \langle p p \rangle$ | trivial, but <b>must be kept</b> |

Note the last row: the current-algebra commutator converts a vanishing MV correlator into a surviving one. Dropping  $[p, p]$  is therefore **not** a harmless simplification at this order. (The paper's Eq. (4.28)–(4.29) does exactly this decomposition.)

## 2.5 Ordinary moments

$$\begin{aligned}
\langle N \rangle &= F_1 \\
\langle N^2 \rangle &= F_1 + F_2 \\
\langle N^3 \rangle &= F_1 + 3F_2 + F_3 \\
\langle N^4 \rangle &= F_1 + 7F_2 + 6F_3 + F_4
\end{aligned}$$

(Stirling numbers of the second kind:  $\langle N^p \rangle = \sum_j S(p, j) F_j$ .)

**Do not quote ordinary moments as results.** They are dominated by trivial powers of  $\langle N \rangle$  —  $\langle N^4 \rangle \approx \langle N \rangle^4$  to a part in  $10^4$  — so all the physics sits in a cancellation of four leading digits. Publish  $C_p$ , not  $\langle N^p \rangle$ . This is also the numerically stable choice (Part VI).

## 2.6 What is complete and what is not — read this before proposing the paper

| Quantity                        | Leading order | Complete with the $0(g^2)$ $\Omega$ ? |
|---------------------------------|---------------|---------------------------------------|
| $\langle N \rangle$             | $0(g^2)$      | ✓ complete through $0(g^4)$           |
| $C_2$ (variance beyond Poisson) | $0(g^4)$      | ✓ <b>complete</b>                     |
| $C_3$                           | $0(g^6)$      | ✗ <b>incomplete</b>                   |
| $C_4$                           | $0(g^8)$      | ✗ <b>incomplete</b>                   |

**Why  $C_3$  is incomplete.**  $\Omega|0\rangle$  has no three-gluon component at  $0(g^2)$  (Part I §1.5), so the three-gluon amplitude first arises at  $0(g^3)$  — and it contributes to  $C_3$  at  $0(\alpha^{*3} T) = 0(g^3 \cdot g^3) = 0(g^6)$ , i.e. *the same order* as the Gaussian contribution  $w^- K^2 w$ . The  $0(g^2)$  operator simply does not contain it. By contrast  $C_2$  is safe: a cubic amplitude first corrects  $m$  at  $0(g^4)$ , which is

subleading to  $m = B_2 = 0(g^2)$ .

**Consequences for the project.** You can honestly publish:

- $\langle N \rangle$  to  $0(g^4)$ , with the JIMWLK-exponentiation check;
- $C_2$  (equivalently the variance) **complete** at  $0(g^4)$  — this is the headline;
- $C_3, C_4$  **only** in their classical colour-fluctuation part (§2.7), which is complete at its own leading order because it needs nothing but the LO displacement;
- the *Gaussian part* of  $C_3, C_4$  as a partial result, clearly labelled as such.

You cannot honestly publish "the third and fourth cumulants at NLO". A referee who knows LCPT will spot the missing three-gluon amplitude immediately. **Either restrict the claim to  $C_2$ , or extend  $\Omega$  to  $0(g^3)$  first** — see Part VIII, Project B.

## 2.7 The colour average, and why it does not rescue the higher cumulants (but is still needed)

Apply the law of total cumulance (§1.8). The  $\kappa_p[\mu]$  terms — cumulants of the *fixed-p mean*  $\mu(p) = \langle N \rangle_p$  over the CGC ensemble — need only the LO displacement, so they are complete at their leading order for every  $p$ . In MV,

$$\mu(p) = \int_k (g^2/k^2) p^a(k) p^a(-k) / (2\pi)^2 \rightarrow \kappa_p[\mu] \propto g^{2p} \cdot (\text{contractions of } p^{2p})$$

and each connected cumulant is suppressed by one power of the number of degrees of freedom  $N_{\text{dof}} \approx (N_c^2 - 1) S_\perp Q_s^2$  per order:

$$\kappa_p[\mu] / \mu^p \sim 1 / N_{\text{dof}}^{p-1}$$

**Crucial and uncomfortable fact:** the classical term  $\kappa_p[\mu]$  and the quantum term  $E_p[C_p^{\text{Fock}}]$  are **the same order in  $g$**  and carry **the same  $1/N_{\text{dof}}^{p-1}$  suppression**. They cannot be separated by power counting. Anyone claiming "we computed the new quantum fluctuation" must show a *structural* discriminator. There is one, and it is good:

**Momentum-space signature.** Bose enhancement (the classical/ $n$ -type correlation) peaks at  $k \approx +q$  — a near-side, same-momentum enhancement. Pair emission from the squeezing kernel ( $m$ -type) peaks at  $k \approx -q$  — **back-to-back**, because the two gluons are emitted from the same vertex and share the source recoil. Plotting the two-gluon correlation as a function of the angle between  $k$  and  $q$  separates them cleanly.

That observation is, I think, the strongest single piece of new physics available in this program, and it survives the "your effect is swamped" objection because the two effects live at different angles rather than at different magnitudes.

## Summary of Part II

- Use  $G(z) = \det[1 - (z-1)K]^{-1/2} \exp\{\dots\}$  — exact at fixed  $p$ , verified against four quantum-optics limits.
- $C_p = (p-1)!/2 \cdot \text{Tr}(K^p) + p!/2 \cdot w^T K^{p-1} w$ , so  $C_p = 0(g^{2p})$ .
- $\langle N \rangle$  complete to  $0(g^4)$ ;  $C_2$  complete at  $0(g^4)$ ;  $C_3, C_4$  **not** complete without the  $0(g^3)$  coefficients of  $\Omega$ .
- The  $0(g^3)$  term in  $\langle N \rangle$  needs  $\langle ppp \rangle$  and vanishes in MV — an odderon-sensitive observable.
- Classical and quantum fluctuations are degenerate in  $g$  and in  $1/N_{\text{dof}}$ ; separate them by the near-side ( $k \approx +q$ ) vs back-to-back ( $k \approx -q$ ) momentum structure.

# Part III — Connected correlations and cumulants

## 3.1 Conversion table

With  $\kappa_n = \sum_p S(n, p) C_p$  (Stirling numbers of the second kind):

$$\begin{aligned} \kappa_1 &= C_1 \\ \kappa_2 &= C_1 + C_2 \\ \kappa_3 &= C_1 + 3C_2 + C_3 \\ \kappa_4 &= C_1 + 7C_2 + 6C_3 + C_4 \\ \kappa_5 &= C_1 + 15C_2 + 25C_3 + 10C_4 + C_5 \end{aligned}$$

Read these as: **a Poisson distribution has  $C_p = 0$  for  $p \geq 2$ , hence  $\kappa_n = \langle N \rangle$  for all  $n$** . Every deviation of  $\kappa_n$  from  $\langle N \rangle$  is a factorial cumulant, i.e. a genuine correlation.

This is why the factorial cumulants  $C_p$  — not  $\kappa_n$  — are the right observables here. They are the *background-free* quantities: they are exactly zero for the leading-order coherent state, so any nonzero value is unambiguously an NLO or a colour-fluctuation effect. Reporting  $\kappa_4$  instead means reporting a number whose leading contribution is  $\langle N \rangle$  and whose physics is a per-mille correction on top.

**Recommendation: make  $C_2$  the headline observable of the follow-up paper, and quote  $\kappa_2 = \langle N \rangle + C_2$  only for contact with the heavy-ion cumulant literature.**

## 3.2 The cumulants, order by order

### $\kappa_2$ — variance

$$\begin{aligned} \kappa_2 &= \langle N \rangle + C_2 \\ &= \underbrace{\langle N \rangle}_{\text{LO}} + \underbrace{\left[ 2\text{Re} \int A_2^* B_2 A_2^* + \int |B_2|^2 \right]}_{\text{quantum, } 0(g^4), \text{ NEW}} \underbrace{\rho\text{-averaged}}_{\text{classical, } 0(g^4), \text{ known}} + \text{Var}_\rho[\mu(\rho)] \end{aligned}$$

- $\langle N \rangle$  (**Poisson part**). Present at LO. Trivially the coherent-state result. Not new.
- $\text{Var}_\rho[\mu]$ . Colour-charge fluctuations of the classical WW field. This is the CGC origin of the negative binomial distribution (Gelis–Lappi–McLerran "Glittering Glasmas"). Known.
- $2\text{Re}[A_2^* B_2 A_2^*] + \int |B_2|^2$ . The genuinely new  $0(g^4)$  piece. Note the first term is an **interference** and therefore **not sign-definite** — the fixed-source distribution can be driven sub-Poissonian. That would be a striking and quotable result, because every existing CGC multiplicity prediction is super-Poissonian (NBD). If a region of  $k_\perp$  or  $Q_s$  gives  $C_2 < 0$ , that is a headline.

**Physical meaning of  $\kappa_2$ :** event-by-event fluctuation of the soft gluon number at fixed valence configuration plus fluctuation of the valence configuration itself.  $\kappa_2/\langle N \rangle = 1$  exactly for a coherent state;  $> 1$  for Bose-enhanced/thermal-like emission;  $< 1$  for destructive displacement–squeezing interference (a genuinely non-classical, "sub-Poissonian light" situation with no analogue in the classical CGC).

### $\kappa_3$ — skewness

$$\kappa_3 = \langle N \rangle + 3C_2 + C_3$$

- $C_3$  is  $0(g^6)$  and **incomplete** with the  $0(g^2)$  operator (Part II §2.6): the missing three-gluon amplitude contributes at the same order.
- The  $3C_2$  term is complete at  $0(g^4)$  and is not negligible — in fact for  $N_{\text{dof}} \gg 1$ ,  $3C_2 \gg C_3$ , since  $C_p/\langle N \rangle^p \sim N_{\text{dof}}^{1-p}$ . So  $\kappa_3$  is **numerically dominated by pieces you can compute completely**, even though  $C_3$  itself is not.

This is a legitimate and defensible publication strategy: *quote  $\kappa_3$  complete through  $0(g^4)$ , i.e.  $\kappa_3 = \langle N \rangle + 3C_2 + 0(g^6)$ , and state that the  $0(g^6)$  remainder is suppressed by  $1/N_{\text{dof}}$* . That is honest and it is a real result.

- The classical part  $\kappa_3^{\text{cl}}[\mu]$  (third cumulant of the WW field over the colour ensemble) is complete and computable; in MV it is the  $\langle p^6 \rangle$  connected topology.



**Physical meaning of  $\kappa_3$ :** asymmetry of  $P(N)$ . Positive skew is the hallmark of a fluctuating-source (compound Poisson / NBD) picture: rare large- $Q_s$  configurations produce a long high-multiplicity tail. A measured/computed skew smaller than the NBD value would indicate that quantum pair correlations partially cancel the source fluctuations.

#### $\kappa_4$ — kurtosis

$$\kappa_4 = \langle N \rangle + 7C_2 + 6C_3 + C_4$$

Same structure: complete through  $O(g^4)$  ( $\langle N \rangle + 7C_2$ ), incomplete at  $O(g^6)$  and beyond.

**Physical meaning of  $\kappa_4$ :** tail weight. In heavy-ion phenomenology  $\kappa_4$  and the ratios  $\kappa_4/\kappa_2$ ,  $\kappa_3/\kappa_2$  are the standard tools for identifying non-trivial correlation structure (they are the same observables used for baryon-number fluctuations and critical-point searches). A CGC-based prediction for the initial-state  $\kappa_4/\kappa_2$  of gluons is a genuinely useful input to that discussion, because it fixes the non-thermal baseline.

### Higher cumulants

$C_p$  for  $p \geq 5$  are all  $O(g^{2p})$  and increasingly incomplete. They are **not** worth computing from this operator. The only exception: the purely classical tower  $\kappa_p^{\text{cl}}[\mu]$  (colour fluctuations of the LO WW field) can be computed to all  $p$  with no extra effort once you sample  $p$  on a lattice — you get the whole distribution for free. Report that as a distribution, not as a list of cumulants.

### 3.3 Which cumulants are nonzero at leading order

| Object               | LO (coherent state, fixed $p$ ) | LO + colour average                | NLO ( $O(g^4)$ )                   |
|----------------------|---------------------------------|------------------------------------|------------------------------------|
| $\langle N \rangle$  | $\int  A_2 ^2 \neq 0$           | $\neq 0$                           | correction                         |
| $C_2$                | 0                               | $\text{Var}_p[\mu] \neq 0$         | first genuine quantum contribution |
| $C_3$                | 0                               | $\kappa_3^{\text{cl}}[\mu] \neq 0$ | partial                            |
| $C_4$                | 0                               | $\kappa_4^{\text{cl}}[\mu] \neq 0$ | partial                            |
| $\kappa_2$           | $= \langle N \rangle$ (Poisson) | $> \langle N \rangle$ (NBD)        | $\approx$                          |
| $\kappa_3, \kappa_4$ | $= \langle N \rangle$           | $> \langle N \rangle$              | $\approx$                          |

#### The single sharpest statement in the whole program:

At leading order and at fixed valence colour charge, the soft gluon multiplicity distribution is exactly Poissonian: all factorial cumulants vanish. The first departure from Poisson statistics arises at  $O(g^4)$  and is controlled entirely by the pair-emission (squeezing) coefficient  $B_2$  of the NLO evolution operator, interfering with the classical Weizsäcker–Williams displacement  $A_2$ .

That is a clean, falsifiable, one-sentence abstract for a paper. It is also exactly what the Introduction of arXiv:2607.18373 promises to deliver.

### 3.4 What contains information beyond the coherent-state approximation

Rank-ordered by how much is genuinely beyond LO:

1. **The anomalous correlator  $m = \langle aa \rangle_c = B_2$ .** Zero for any coherent state. It is the *only*  $O(g^2)$  object in the theory that is invisible at LO. Everything new in  $C_2$  traces back to it.
2. **The back-to-back structure of  $m_{\{k, q\}}$**  (peaked at  $q \approx -k$ ). No coherent state and no Bose-enhancement mechanism produces this. See Part IV §4.3 — this is the discriminator.
3. **The sign of  $C_2$ .** Coherent  $\rightarrow 0$ ; classical fluctuating source  $\rightarrow$  strictly  $> 0$ ; quantum interference  $\rightarrow$  can be  $< 0$ . A negative region is unreachable by any classical CGC calculation.

4. **The  $\langle ppp \rangle$  (odderon) contribution to  $\langle N \rangle$  at  $O(g^3)$ .** Vanishes in MV; nonzero with a C-odd weight. This is orthogonal to everything above and is essentially virgin territory.
5.  $n = B_2^\dagger B_2$  — the true "pair occupancy". Beyond LO but  $O(g^4)$  in the *kernel*, hence  $O(g^8)$  in  $C_2$ . Numerically irrelevant. Do not build a project on it.

### 3.5 The one thing that will get the paper into trouble

**Gluon number in the light-cone wave function is not an observable.** It is gauge-dependent (defined in  $A^+=0$ ), scheme-dependent (depends on the  $k^+$  cutoff separating soft from valence — the paper's  $\Lambda$  and  $v$ ), and IR-divergent. A referee will ask, correctly:

*"You have computed the fluctuations of a number operator in a particular gauge and a particular factorization scheme. What is measured?"*

There are three defensible answers, in increasing order of work and of persuasiveness:

- **(a) It is an intermediate quantity.** Present  $C_2$  as a property of the CGC wave function, analogous to how the unintegrated gluon distribution is presented. Acceptable, but weak, and will draw a "limited physics impact" report.
- **(b) Show that the cutoff dependence is JIMWLK.** Demonstrate that  $\partial \langle N \rangle / \partial Y$  and  $\partial C_2 / \partial Y$  are generated by the JIMWLK Hamiltonian acting on the lower-order results. This makes the scheme dependence *physics* rather than an artefact. **Strongly recommended; cheap.**
- **(c) Compute produced-gluon cumulants.** Scatter the dressed state off a target: insert Wilson lines,  $a \rightarrow U a$ , and compute the multiplicity of *produced* gluons. This converts the whole calculation into an observable. It roughly doubles the work (you need  $\langle U U U U \rangle$ -type target correlators, i.e. quadrupoles) but it is what turns a technical note into a physics paper — and it is the natural consumer of the paper's  $C, D, E$  coefficients, which play no role in  $\Omega|0\rangle$ .

**My assessment: (b) is mandatory, (c) is what separates a PRD from a JHEP.** See Part VIII.

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## Summary of Part III

- $C_2$  is the right headline observable: exactly zero at LO, complete at  $O(g^4)$ , sign-indefinite.
- $K_3, K_4$  are complete only through  $O(g^4)$  ( $\langle N \rangle + 3C_2, \langle N \rangle + 7C_2$ ); their  $O(g^6)$  pieces need the  $O(g^3)$  operator. Say so explicitly rather than hoping nobody checks.
- Everything genuinely new traces to the anomalous correlator  $m = B_2$ , and its distinguishing signature is back-to-back momentum structure.
- The wave-function-vs-observable problem is the paper's main vulnerability; fix it with the JIMWLK consistency check at minimum.

# Part IV — Correlation functions

## 4.1 Single-gluon distribution

$$\begin{aligned} dN/d^2k &\equiv \sum_{\{a,i\}} \langle \Psi | a^\dagger_{a\uparrow i}(k) a_{a\downarrow i}(k) | \Psi \rangle \\ &= \sum_{\{a,i\}} [ |\alpha_{a\downarrow i}(k)|^2 + n_{aa\downarrow i}(k,k) ] \end{aligned}$$

At LO this is the Weizsäcker–Williams / unintegrated gluon distribution:

$$dN/d^2k|_{LO} = (g^2/k^2) \langle \rho_a(k) \rho_a(-k) \rangle / (2\pi)^2 \sim (N_c^2-1) g^2 \mu^2 S_\perp / k^2$$

with the familiar  $1/k^2$  tail and IR log divergence cut off by saturation at  $k \sim Q_s$ . Nothing new; it is the normalization against which everything else is measured.

The  $O(g^4)$  correction  $n(k,k) = (B_2 + B_2')(k,k)$  is the **pair-production contribution to the gluon spectrum**. It is the real-emission part of the NLO unintegrated gluon distribution and it is worth plotting on its own: it has a *different*  $k$ -dependence from the LO term (harder, because pair emission from the  $g \rightarrow gg$  vertex populates larger  $k$ ), so the ratio  $n(k,k) / |\alpha(k)|^2$  rises with  $k_\perp$ .

## 4.2 Two-gluon correlation

$$\begin{aligned} dN_2/d^2k d^2q &= \langle a^\dagger_k a^\dagger_q a_k a_q \rangle \\ &= (dN/d^2k)(dN/d^2q) + \Gamma(k,q) \end{aligned}$$

The connected piece, from the Gaussian Wick expansion of §2.2:

$$\begin{aligned} \Gamma(k,q) &= |n(k,q)|^2 + |m(k,q)|^2 \\ &\quad + 2 \operatorname{Re} [ \alpha^*(k) n(k,q) \alpha(q) ] \quad \leftarrow \text{Bose enhancement / HBT, peaks at } k \approx +q \\ &\quad + 2 \operatorname{Re} [ \alpha^*(k) \alpha^*(q) m(k,q) ] \quad \leftarrow \text{pair emission, peaks at } k \approx -q \end{aligned}$$

with the orders

$$\begin{array}{lll} |m(k,q)|^2 & O(g^4) & \\ 2\operatorname{Re}[\alpha^* \alpha^* m] & O(g^4) & \leftarrow \text{LEADING, sign-indefinite} \\ 2\operatorname{Re}[\alpha^* n \alpha] & O(g^6) & \\ |n(k,q)|^2 & O(g^8) & \end{array}$$

**This is the central result of Part IV.** At  $O(g^4)$  the connected two-gluon correlation of the NLO wave function is

$$\Gamma(k,q)|_{O(g^4)} = |B_2(k,q)|^2 + 2 \operatorname{Re} [ A_2^*(k) A_2^*(q) B_2(k,q) ]$$

## 4.3 Why the angular structure is the discriminator

Both known CGC mechanisms and the new one produce a connected two-gluon correlation of the same parametric size ( $\sim (N)^2/N_{\text{dof}}$ ), so magnitude alone proves nothing. The angular structure does:

| Mechanism                              | Correlator                   | Peaks at                                              | Origin                                                                                      |
|----------------------------------------|------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Bose enhancement in projectile WF      | $\alpha^* n \alpha$          | $k \approx +q$ (near side, $\Delta\phi \approx 0$ )   | identical-boson symmetrization; the Altinoluk–Armesto–Beuf–Kovner–Lublinsky ridge mechanism |
| HBT                                    | $\alpha^* n \alpha$          | $k \approx +q$                                        | same object, different kinematic limit                                                      |
| <b>Pair emission / squeezing</b>       | $\alpha^* \alpha^* m$        | $k \approx -q$ (away side, $\Delta\phi \approx \pi$ ) | both gluons from one vertex; transverse momentum conservation against the source            |
| Colour-charge (classical) fluctuations | enters via $\text{Var}_\rho$ | broad, $\Delta\phi$ -independent                      | fluctuating $Q_s$                                                                           |

So: **plot  $\Gamma(k,q)$  versus  $\Delta\varphi$  at fixed  $|k| = |q|$** . The  $\mathcal{O}(g^4)$  NLO term contributes an away-side ridge that the glasma-graph/Bose-enhancement calculations do not produce. That is a concrete, checkable, publishable prediction, and it connects directly to the ridge literature.

**Honest caveat:** an away-side correlation from momentum conservation is *not* by itself surprising, and referees will point out that momentum-conservation-induced away-side correlations are ubiquitous and often treated as background. The claim must therefore be quantitative — the *normalization and  $Q_s$ -dependence* of the away-side term, not its existence.

## 4.4 Three-gluon correlation

$$dN_3/d^2k_1d^2k_2d^2k_3 = \langle a^\dagger_1 a^\dagger_2 a^\dagger_3 a_1 a_2 a_3 \rangle$$

Connected part, at leading order in the kernels:

$$\Gamma_3(k_1, k_2, k_3) \big|_{\text{leading}} = 2 \operatorname{Re} [ \alpha^*_{12} \alpha^*_{13} m_{12} \cdot (\alpha^*_{23} \dots) ]\text{-type} + \operatorname{Re} [ m_{12} n_{23} \dots ] \\ \sim \mathcal{O}(g^6)$$

**Incomplete** for the same reason  $C_3$  is: an  $\mathcal{O}(g^3)$  three-gluon amplitude in  $\Omega|0\rangle$  contributes a genuine three-gluon emission vertex at  $\mathcal{O}(g^6)$ . Physically that is the  $1 \rightarrow 3$  splitting, and it is *precisely* the object one would expect to dominate a three-gluon correlation.

**Recommendation: do not build a project around the three-gluon correlation from this operator.** The one honest three-gluon statement available is the *Gaussian prediction*: with  $\Omega|0\rangle$  truncated at two-gluon components,  $\Gamma_3$  is completely determined by  $\Gamma_2$  (no independent three-body correlation). Stating and testing that — "the NLO wave function predicts Gaussian-state three-body factorization up to  $\mathcal{O}(g^6)$ " — is a legitimate, if modest, result, and it is a genuinely useful null prediction for the ridge community.

## 4.5 Factorial moments and normalized cumulants

|                                            |                                                            |
|--------------------------------------------|------------------------------------------------------------|
| $F_p = \langle N(N-1)\dots(N-p+1) \rangle$ | factorial moments                                          |
| $C_p$                                      | factorial cumulants (§2.2)                                 |
| $f_p = F_p / \langle N \rangle^p$          | normalized factorial moments $\rightarrow 1$ for Poisson   |
| $c_p = C_p / \langle N \rangle^p$          | normalized factorial cumulants $\rightarrow 0$ for Poisson |

$c_p$  is the correct dimensionless observable. Expected scaling:

$$c_p \sim 1 / N_{\text{dof}}^{p-1}, \quad N_{\text{dof}} \approx (N_c^2 - 1) S_{\perp} Q_s^2 / (2\pi)^2$$

so  $c_2 \sim 1 / [(N_c^2 - 1) S_{\perp} Q_s^2]$ . **This is the same  $1/k$  that appears in the negative binomial fits to pp and AA multiplicity data**, and connecting the NLO calculation to it is the most direct route to phenomenological relevance:

$k_{\text{NBD}} = 1/c_2$  — computing  $c_2$  from the NLO wave function is computing the NBD parameter  $k$  from first principles, including its  $\mathcal{O}(g^4)$  quantum correction.

That framing is much more likely to be read and cited than "we computed a factorial cumulant".

## 4.6 Generating function

Fixed  $p$  (exact through  $\mathcal{O}(g^2)$ , from §2.2):

$$G_p(z) = \det[1 - (z-1)K]^{-1/2} \exp\{ \frac{1}{2}(z-1) w^T [1 - (z-1)K]^{-1} w \}$$

Full distribution:

$$G(z) = \int Dp \, W_Y[p] \, G_p(z) \quad P(N) = (1/N!) \, d^N G/dz^N \big|_{z=0}$$

Three tractable evaluations:

1.  $K = 0$  (LO):  $G_p(z) = e^{\{(z-1)\mu(p)\}}$ , so  $G(z) = \int Dp W[p] e^{\{(z-1)\mu(p)\}}$  — a **compound Poisson** distribution. If  $\mu(p)$  were Gamma-distributed this gives exactly the negative binomial. Deriving the actual  $\mu$ -distribution in MV and showing how close it is to Gamma is a clean, self-contained calculation and directly connects to the "KNO scaling from a nearly Gaussian action" line of work.
2.  $K \neq 0$ , **leading order**: expand  $\log G_p$  to  $O(g^4)$ ;  $P(N)$  is a Poisson deformed by a single factorial-cumulant correction — analytic, and gives the shape distortion in closed form.
3. **Numerical**: sample  $p$ , build  $K$  and  $w$ , evaluate the Fredholm determinant on a momentum grid, invert the  $z$ -transform by FFT. Gives full  $P(N)$  including tails. See Part VI.

**KNO scaling** is directly accessible:  $\Psi(N/\langle N \rangle) = \langle N \rangle P(N)$ . Whether KNO scaling survives the  $O(g^4)$  correction — i.e. whether  $c_p$  are  $\langle N \rangle$ -independent — is a well-posed, interesting, and to my knowledge unanswered question. **KNO holds iff  $c_p$  depends on  $Y$  and  $Q_s$  only through  $\langle N \rangle$** . Since  $c_2 \sim 1/(S_{\perp Q_s^2})$  and  $\langle N \rangle \sim S_{\perp Q_s^2}/\alpha_s$ , we get  $c_2 \sim \alpha_s/\langle N \rangle$  — so KNO is **violated by a running- $\alpha_s$  factor** at LO, and the  $O(g^4)$  term modifies this. That is a sharp, quotable prediction and it is testable against the existing KNO literature.

## 4.7 Which observables are worth the effort

| Observable                                | Theory interest     | Experimental contact                  | Verdict                  |
|-------------------------------------------|---------------------|---------------------------------------|--------------------------|
| $dN/d^2k$ at $O(g^4)$                     | medium (NLO ugd)    | indirect                              | do it, it is a byproduct |
| $c_2 = 1/k_{\text{NBD}}$                  | <b>high</b>         | <b>direct</b> (NBD fits, $pp/pA/AA$ ) | <b>primary target</b>    |
| $\Gamma(k, q)$ vs $\Delta\phi$            | <b>high</b>         | direct (ridge, away-side)             | <b>primary target</b>    |
| KNO violation from $c_p$                  | high                | direct (H1, ALICE, CMS)               | strong secondary         |
| $\Gamma_3$                                | low (incomplete)    | weak                                  | null prediction only     |
| $\langle N \rangle$ at $O(g^3)$ , odderon | medium-high (novel) | weak (needs C-odd observable)         | interesting side project |
| $K_4, K_4/K_2$                            | medium              | contact with fluctuation programme    | secondary                |

## Summary of Part IV

- Connected two-gluon correlation at  $O(g^4)$ :  $\Gamma = |B_2|^2 + 2\text{Re}[A_2^* A_2 B_2]$ .
- Its angular signature is **away-side** ( $k \approx -q$ ), distinguishing it from Bose enhancement and HBT, which are near-side. This is the cleanest new prediction available.
- Normalized factorial cumulant  $c_2 = 1/k_{\text{NBD}}$ : a first-principles NLO computation of the negative binomial parameter. Best route to phenomenological impact.
- The generating function is a Fredholm determinant at fixed  $p$ , compound-Poisson after colour averaging; KNO scaling is violated at the level of  $\alpha_s$ , computably.
- Three-gluon correlations are not honestly computable here — publish the Gaussian-factorization null prediction instead.

## Part V — Model calculations before the full CGC average

Ordered from "do this in an afternoon" to "this is the actual paper". For each: what to derive, what the integrals are, the cost, and what gets plotted.

Throughout,  $A_2$  and  $B_2$  are the paper's Eqs. (5.2) and (5.3).

---

### Model 1 — Fixed classical source (single configuration)

**Setup.**  $\rho^a(x)$  a fixed c-number field; no averaging at all.  $[\rho, \rho] \rightarrow 0$ .

**Derive.**

$$\begin{aligned}\alpha_k &= A_2, k[\rho] = i g (k_i/k^2) \rho^a(k) \times (\text{norm}) \\ m_{\{kq\}} &= B_2, \{kq\}[\rho] \\ C_2 &= 2\text{Re} \int_{\{k, q\}} A_2^* \cdot_k B_2, \{kq\} A_2^* \cdot_q + \int_{\{k, q\}} |B_2, \{kq\}|^2\end{aligned}$$

Everything is a deterministic functional.  $P(N)$  from the Fredholm determinant.

**Integrals.** Two 2-D transverse integrals  $\Rightarrow$  **4-D**. With colour and polarization sums: a 4-D quadrature times  $O((N_c^2-1)^2 \cdot 4)$  index terms.

**Cost.** Seconds. Adaptive quadrature or a  $256^2$  grid.

**Plot.**  $P(N)$  versus  $N$  for a single "event", overlaid with the Poisson of the same mean. The message: *how far from Poisson does one configuration get?* This is the cleanest possible display of the new physics because there is no colour averaging to hide behind.

**Value.** This is the single most important sanity plot in the whole program and it is nearly free. **Do it first.** If  $C_2$  comes out numerically negligible for a realistic single configuration, the project's headline claim is in trouble and you want to know that on day one.

---

### Model 2 — Toy one-dimensional source

**Setup.** Replace  $d^2k \rightarrow dk, 1/k^2 \rightarrow 1/k$ , one colour. An Abelian scalar toy.

**Derive.** Everything analytically.  $A_2(k) = g \rho(k)/k$ , choose  $B_2(k, q) = g^2 \rho(k+q) f(k, q)/(kq)$  for a model splitting kernel  $f$ .

**Integrals.** 2-D. Often closed form.

**Cost.** Instant; much of it symbolic.

**Plot.**  $C_2$  versus the IR cutoff;  $C_2$  versus source width. Demonstrates the sign structure of the interference term without any colour algebra.

**Value.** High as a *debugging and intuition* tool, zero as a publishable result. Use it to verify the Fredholm-determinant code against exact answers before turning on 2-D.

---

### Model 3 — Finite number of point colour charges

**Setup.**  $\rho^a(x) = \sum_{n=1}^{N_s} g T^a_n \delta^2(x - x_n)$ , with  $N_s$  charges at fixed or random positions and random colour orientations.

**Derive.**

$$A_{2,i}^a(k) = i g \sum_n (k_i/k^2) T^a_n e^{-ik \cdot x_n}$$

The  $1/N_s$  suppression of connected correlations becomes explicit:  $c_2 \propto 1/N_s$  up to interference phases.

**Integrals.** Same 4-D, but the source integrals are replaced by finite sums  $\Rightarrow$  **the transverse integrals become analytic** for point sources, leaving only sums over  $n$ .  $O(N_s^4)$  colour terms for  $C_2$ .

**Cost.**  $N_s \leq 100$ : milliseconds.  $N_s = 10^4$ : still trivial with FFT.

**Plot.**  $c_2 \cdot N_s$  versus  $N_s$  — should approach a constant, verifying the  $1/N_{\text{dof}}$  scaling law. Also  $c_2$  vs  $N_s$  at fixed total charge, separating "more sources" from "stronger sources".

**Value.** This is the cleanest demonstration that the effect scales as advertised, and it pre-empts a referee asking "is your effect just  $1/N$ ?". **Include it as a figure.**

---

## Model 4 — MV / Gaussian colour charge distribution

**Setup.**  $W[p] \propto \exp\{-\int d^2x \, p^a(x) p^a(x) / (2 g^2 \mu^2)\}$ , so  $\langle p^a(x) p^b(y) \rangle = g^2 \mu^2 \delta^{ab} \delta^2(x-y)$ .

**Derive.** Two equivalent routes:

**(4a) Analytic Wick reduction.** Every correlator factorizes into pairings. For  $C_2$  you need  $\langle p^4 \rangle = 3$  pairings. Each pairing produces a distinct transverse convolution. Schematically:

$$\langle 2 \text{Re} \int A_z^* \cdot_k B_z, \{kq\} A_z \cdot_q \rangle \rightarrow g^6 \mu^4 S_{\perp} \int d^2k d^2q \, (k_i q_j / k^2 q^2) \theta_{\{ij\}}(k,q) \times [\text{colour factor}]$$

with  $\theta$  the  $p$ -stripped splitting kernel. Colour factors:  $f^{abc} f^{abc} = N_c(N_c^2 - 1)$ ,  $\text{Tr}(T^a T^b) = \dots$  — a finite table,  $\sim 10$  entries.

**Integrals.** After the delta functions from  $\langle pp \rangle$  are used,  $C_2$  reduces to **one or two 2-D integrals  $\Rightarrow$  2-D to 4-D**.  $C_3$  would be 4-D to 8-D;  $C_4$  up to 12-D.

**Cost.**  $C_2$ : adaptive 4-D quadrature, seconds to minutes; or Vegas with  $10^6$  points.  $C_3, C_4$ : Vegas with  $10^7$ – $10^8$  points, hours — and with severe cancellation, so error bars will be the limiting factor. **This is why I recommend route (4b) for anything above  $C_2$ .**

**(4b) Direct lattice sampling — the recommended production method.**

```
for each of N_conf configurations:
  sample p^a(x) as Gaussian white noise on an LxL transverse lattice
  FFT  $\rightarrow$  p^a(k)
  build  $\alpha(k) = A_z[p]$ , and the kernel  $m(k,q) = B_z[p]$ 
  compute  $\mu = \Sigma |\alpha|^2$ ,  $C_2[p]$  from the  $O(g^4)$  formula
  accumulate the ensemble cumulants of the full P(N) via the law of total cumulance
```

**Cost.**  $O(N_{\text{conf}} \cdot L^2 \log L)$  for the displacement; the kernel  $m(k,q)$  is the bottleneck at  $O(L^4)$  if stored densely. Mitigations: (i)  $m$  is a convolution in  $k+q$  for the  $pp$  piece, so it is  $O(L^2 \log L)$  with FFT; (ii) evaluate  $\text{Tr}(K^p)$  stochastically with Hutchinson trace estimators (random probe vectors) rather than forming  $K$  — this is standard in lattice QCD and reduces the cost to  $O(N_{\text{probe}} \cdot L^2 \log L)$ .

With  $L = 512$ ,  $N_{\text{conf}} = 10^5$ ,  $N_{\text{probe}} = 20$ : **a few CPU-hours, or minutes on a GPU**. This is entirely feasible for a single author. Higher cumulants need  $N_{\text{conf}} \sim 10^6$ – $10^7$ , still feasible.

**Plot.**  $c_2$  vs  $Q_s$  ( $=g^2 \mu / S_{\perp}$ -ish);  $P(N)$  vs Poisson and vs NBD; KNO function.

**Value.** **This is the paper.** MV is the standard, referees expect it, and it is where the  $1/k_{\text{NBD}}$  connection is made.

---

## Model 5 — Translationally invariant source

**Setup.** Take  $S_{\perp} \rightarrow \infty$  with  $\mu^2$  fixed; all correlators diagonal in momentum,  $\langle p(k) p(q) \rangle = (2\pi)^2 \delta^2(k+q) g^2 \mu^2 S_{\perp}$ .

**Effect.** Each momentum delta function removes 2 dimensions.  $C_2$  collapses from 4-D to **2-D**;  $C_3$  from 8-D to 4-D;  $C_4$  from 12-D to 6-D.

**Cost.**  $C_2$ : milliseconds.  $C_4$ : **6-D Vegas, minutes**. This is the *only* setting in which the fourth cumulant is comfortably computable.

**Caveat.** Translational invariance kills all impact-parameter/geometry fluctuations, which are a large part of the physical  $\text{Var}_p[\mu]$ . So this model gives you the *quantum* part cleanly and throws away the classical part. **That is actually a feature:** it is the ideal setting for isolating the new  $\mathcal{O}(g^4)$  effect. Present it as "the quantum contribution in the translationally-invariant limit", then add geometry in Model 4.

**Plot.**  $c_2, c_3, c_4$  versus  $k_\perp$  and versus  $Q_s$ , with the classical contribution switched off by construction.

## Model 6 — Large $N_c$

**Setup.** Keep only leading  $N_c$  in every colour contraction.

**Effect.** The colour-factor table collapses to a handful of terms;  $f^{\{abc\}}f^{\{abd\}} = N_c \delta^{\{cd\}}$  dominates and quadrupole-type contractions factorize into dipoles.

**Cost.** Reduces the colour bookkeeping by roughly an order of magnitude; no change in integral dimensionality.

**Value.** Essential as a **cross-check**, not as the main result. The  $1/N_c^2$  corrections to  $c_2$  are interesting in their own right (they are  $\mathcal{O}(10\%)$  for  $N_c=3$ ) and a plot of  $c_2(N_c)/c_2(\infty)$  is a cheap extra figure. **Do not present large- $N_c$  as the primary result** — for multiplicity fluctuations the subleading- $N_c$  terms carry the Bose-enhancement structure, and a referee will ask about exactly that.

## Model 7 — Gaussian (non-local) colour charge distribution / MV with $Q_s$ -fluctuations

**Setup.**  $\langle pp \rangle = G(x-y)$  with a nontrivial correlator, or MV with  $\mu^2$  itself fluctuating event-by-event ( $\mu^2 \rightarrow \mu^2(1 + \delta)$  with  $\delta$  Gaussian or log-normal).

**Why.**  $\mu^2$ -fluctuations are the dominant known source of  $k_3, k_4$  in phenomenology and they are *not* in the standard MV weight. Including them is what makes contact with the "fluctuating  $Q_s$ " literature, and it lets you show explicitly that the quantum  $c_2$  survives on top of them.

**Integrals.** One extra 1-D integral over  $\delta$  per observable.

**Cost.** Negligible.

**Plot.**  $c_2^{\text{total}}$  decomposed into quantum and classical parts as a function of the width of the  $Q_s$  fluctuation. **This is your defence against the "your effect is swamped" referee report** — it quantifies exactly when it is swamped and when it is not.

## Complexity summary

| Model                    | $c_2$ dim | $c_3$ dim | $c_4$ dim | Cost for $c_2$ | Cost for $c_4$ |
|--------------------------|-----------|-----------|-----------|----------------|----------------|
| 1. Fixed source          | 4         | 6         | 8         | seconds        | minutes        |
| 2. 1-D toy               | 2         | 3         | 4         | instant        | instant        |
| 3. $N_s$ point charges   | sums      | sums      | sums      | ms             | seconds        |
| 4. MV, analytic Wick     | 4         | 8         | 12        | minutes        | hours-days ✗   |
| 4b. MV, lattice sampling | —         | —         | —         | CPU-hours      | CPU-hours ✓    |
| 5. Translationally inv.  | 2         | 4         | 6         | ms             | minutes ✓      |
| 6. Large $N_c$           | as above  | as above  | as above  | ÷10 colour     | ÷10            |
| 7. $Q_s$ -fluctuating MV | 4+1       | 8+1       | 12+1      | minutes        | via 4b         |

**Recommended sequence:** 2 → 1 → 3 → 5 → 4b → 7 → 6. That order front-loads all the debugging and puts the expensive production run (4b) after every analytic cross-check exists.



## Two technical warnings

1. **The  $k$ -integrals are IR divergent** at LO ( $\int d^2k/k^2$ ). Every quantity must be defined with an explicit IR regulator, and you must show which combinations are IR-finite.  $c_2 = C_2/\langle N \rangle^2$  involves a ratio of divergent quantities — **check whether the divergences cancel in the ratio**. If they do, that is a result worth stating. If they do not,  $c_2$  is cutoff-dependent and must be quoted at fixed  $k_\perp$  rather than integrated. My expectation: the ratio is *not* IR-finite, because  $C_2$  and  $\langle N \rangle^2$  have different IR weights. **Plan for  $k_\perp$ -differential observables from the start.**
2. **Cancellations in  $C_p$ .**  $C_4$  is a difference of quantities of order  $\langle N \rangle^4$ ; naive Monte Carlo loses ~4 digits. Always compute  $C_p$  from the *cumulant* estimator directly (or from the Fredholm determinant), never as a difference of sampled moments.

## Part VI — Proposed numerical studies

For each: why it matters, whether it exists in the literature, and what behaviour to expect. "Novel?" is judged against the survey in Part VII.

### P1. Multiplicity distribution $P(N)$ , fixed single configuration

**Why.** The most direct display of the  $O(g^4)$  effect with nothing averaged away. **In the literature?**  $P(N)$  from CGC exists (Gelis–Lappi–McLerran, Glasma flux tubes) but **always after colour averaging and always at classical order**. A fixed-configuration  $P(N)$  from an NLO wave function does not exist. **Novel. Expect.** Nearly Poisson; deviation at the  $1/N_{\text{dof}}$  level, i.e. a few percent for realistic  $Q_s$ . Plot  $P(N)/P_{\text{Poisson}}(N)$  — a mild S-shape (excess in the tails if  $C_2 > 0$ , depletion if  $C_2 < 0$ ). Do **not** plot  $P(N)$  raw; the deviation is invisible.

### P2. $c_2 = C_2/(N)^2$ versus $Q_s$

**Why.**  $1/c_2$  is the negative-binomial  $k$ . This is the observable with the most direct experimental contact. **In the literature?**  $k_{\text{NBD}}(Q_s)$  exists at classical order ( $k \propto (N_c^2-1)Q_s^2 S_{\perp}/2\pi$ , Gelis–Lappi–McLerran). The  $O(g^4)$  quantum correction to it does **not** exist. **Novel as a correction. Expect.**  $c_2 \approx a/(N_c^2-1)S_{\perp}Q_s^2 \cdot [1 + b \cdot \alpha_s(Q_s) + \dots]$ . The interesting question is the sign and size of  $b$ . Plot  $c_2 \cdot (N_c^2-1)S_{\perp}Q_s^2$  versus  $Q_s$  — should be flat at LO, and its slope *is* the new physics. **This is the money plot of the project.**

### P3. $\kappa_2/(N) - 1$ versus $Q_s$ , decomposed

**Why.** Shows quantum vs classical contributions side by side (law of total cumulance). **In the literature?** The decomposition itself is not in the CGC literature. **Novel. Expect.** Classical part positive and dominant at large  $S_{\perp}Q_s^2$ ; quantum part smaller and possibly of either sign. **If the quantum part is everywhere  $< 1\%$  of the classical part, the project's physics case collapses — this plot is the go/no-go test.** Run it early.

### P4. Cumulants versus rapidity $Y$

**Why.** Tests whether the  $Y$ -dependence is generated by JIMWLK (the consistency check of Part III §3.5b). **In the literature?**  $Y$ -dependence of  $\langle N \rangle$  is standard.  $Y$ -dependence of  $c_2$  from an NLO operator is **novel**. **Expect.**  $\langle N \rangle$  grows like  $e^{\{\omega Y\}}$ ;  $Q_s^2(Y) \propto e^{\{\lambda Y\}}$  with  $\lambda \approx 0.2-0.3$ ; hence  $c_2 \propto e^{\{-\lambda Y\}}$  — **cumulant ratios decrease with rapidity/energy**, i.e. multiplicity distributions narrow (relative to the mean) as energy grows. This is a clean, falsifiable qualitative prediction and it can be confronted with the  $\sqrt{s}$ -dependence of NBD  $k$  in pp data (where  $k$  is observed to *decrease* with  $\sqrt{s}$ , i.e.  $c_2$  *increases* — **note the tension**, and address it: the data are for produced hadrons in a rapidity window, not for the projectile wave function, and the discrepancy is likely a fragmentation/geometry effect. Discuss it; do not hide it).

### P5. Cumulants versus infrared cutoff $\Lambda_{\text{IR}}$

**Why.** Establishes which observables are IR-safe. Non-optional. **In the literature?** Routine in CGC, but not for cumulants. **Novel in this context, low impact. Expect.**  $\langle N \rangle \sim \ln(Q_s/\Lambda_{\text{IR}})$ ;  $C_2$  a different power. Plot  $c_2$  vs  $\Lambda_{\text{IR}}$  — if it plateaus, you have an IR-safe observable and should say so loudly; if it does not, quote everything  $k_{\perp}$ -differentially instead. **Run this before writing any abstract**, because it determines what you are allowed to claim.

### P6. Cumulants versus UV cutoff / $k^+$ (factorization scale)

**Why.** The soft/valence separation is a scheme choice — the paper's  $\Lambda$  and  $v$ ; scheme dependence must be shown to be JIMWLK-compensated. **In the literature?** Standard for  $\langle N \rangle$ . **Novel for  $c_2$ . Expect.** Residual  $\ln v$  dependence at  $O(g^4)$  that cancels against the LO term evolved by one step of JIMWLK. Demonstrating that cancellation numerically is a strong technical result and answers the "your number is scheme-dependent" objection directly. (Note the paper itself raises  $v$ -independence in §2.3 and never returns to it, so this would also close a gap there.)

### P7. $c_2$ versus transverse momentum $k_{\perp}$

**Why.** IR-safe alternative to the integrated cumulant; and the shape carries the discriminating information. **In the literature?**  $k_{\perp}$ -differential multiplicity correlations exist in the glasma-graph literature;  $k_{\perp}$ -differential *factorial cumulants of an NLO wave function* do not. **Novel. Expect.**  $c_2(k_{\perp})$  roughly flat below  $Q_s$ , rising above it, because the pair-emission kernel  $B_2$  is harder than the WW field  $A_2$ . A rise of the correlation strength with  $k_{\perp}$  above  $Q_s$  is a distinctive signature.

## P8. Two-gluon correlation versus $\Delta\phi$ — the discriminator plot

**Why.** Separates the new pair-emission mechanism (away side) from Bose enhancement (near side). See Part IV §4.3. **In the literature?** Near-side ridge from Bose enhancement: Altinoluk–Armesto–Beuf–Kovner–Lublinsky and successors. **An away-side component from the NLO squeezing kernel of the wave function is not in the literature. Most novel item in the list. Expect.**  $\Gamma(\Delta\phi)$  with a near-side peak ( $O(g^6)$ , Bose) and an away-side peak ( $O(g^4)$ , pair emission) — so at  $O(g^4)$  the away-side one should actually **dominate**. That is a surprising and highly quotable statement, and it is the result most likely to get the paper cited outside the small- $x$  formalism community. **Verify it carefully; if true it is the abstract.**

## P9. Dependence on source density $\mu^2 / N_s$

**Why.** Verifies the  $1/N_{\text{dof}}$  scaling and separates "more sources" from "stronger sources". **In the literature?** The scaling law is folklore; an explicit NLO check is **novel but minor**. **Expect.**  $c_2 \cdot N_s \rightarrow \text{const.}$  A deviation signals an error in the calculation. **Use as a code validation plot, put it in an appendix.**

## P10. KNO function $\Psi(z) = \langle N \rangle P(N)$ , $z = N/\langle N \rangle$ , at several $Y$

**Why.** Tests KNO scaling and its violation. **In the literature?** KNO from CGC/dipole models: yes (nearly-Gaussian action; Mueller–Kharzeev–Levin; Le–Mueller–Munier). **KNO from an NLO wave function: no. Novel. Expect.** Approximate KNO with violations of relative size  $\alpha_s(Q_s)$ . Plot  $\Psi(z)$  at three  $Y$  values; the spread between curves *is* the violation.

## P11. $c_3, c_4$ (translationally invariant model only)

**Why.** Completeness; contact with the fluctuation-observable community. **In the literature?** Not for CGC wave functions. **Novel but incomplete** (Part II §2.6). **Expect.**  $c_p \sim N_{\text{dof}}^{\{1-p\}}$ , alternating signs plausible. **Must be labelled "Gaussian contribution only; the  $O(g^6)$  three-gluon amplitude is not included."** If you cannot bring yourself to write that sentence, do not make the plot.

## P12. Sign map of $c_2$ in the $(k_\perp, Q_s)$ plane

**Why.** Locates any sub-Poissonian region — the genuinely non-classical regime. **In the literature?** Nothing comparable. **Novel. Expect.** Most of the plane super-Poissonian; possibly a sub-Poissonian island at large  $k_\perp$  where the interference term dominates over  $|B_2|^2$ . If such a region exists, this becomes the paper's most striking figure — "the CGC wave function is locally sub-Poissonian", with a direct quantum-optics reading (non-classical light).

## Priority ordering for actually doing the numerics

| Rank | Plot    | Purpose                                                |
|------|---------|--------------------------------------------------------|
| 1    | P3      | go/no-go: is the quantum effect big enough to publish? |
| 2    | P5      | determines what can be claimed (IR safety)             |
| 3    | P2      | the money plot: $k_{\text{NBD}}$ at NLO                |
| 4    | P8      | the discriminator: away-side vs near-side              |
| 5    | P7, P12 | $k_\perp$ structure and sign map                       |
| 6    | P4, P6  | JIMWLK/scheme consistency                              |
| 7    | P1, P10 | distributions and KNO                                  |
| 8    | P9, P11 | validation and completeness                            |

**Run P3 and P5 before committing to the project.** They are cheap (Models 1 and 5, hours of work) and they decide whether the physics case survives.

## Part VII — Literature survey: what exists, what does not

**Method note.** This survey was assembled from bibliographic search only (full texts were not retrievable in the session where it was written). Titles, authors, and identifiers are reliable; my characterization of each paper's *detailed* content is inference from title/abstract/known context and should be verified by reading before any of it is repeated in a publication. Items marked (!) **READ THIS FIRST** are ones where the overlap risk is high enough that a wrong guess would waste months.

### 7.1 The direct lineage — NLO light-cone wave functions and evolution operators

| Work                                                                                                                                                                                                                                                                                                    | Content                                                                                                          | Relation to this program                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kovner, Lublinsky, Mulian, <i>JIMWLK at NLO</i> (2013–14)                                                                                                                                                                                                                                               | The $0(g^3)$ /two-loop soft wave function; extraction of $H\_JIMWLK^{NLO}$ from the diagonal $S$ -matrix element | <b>The direct ancestor.</b> They compute the soft wave function to the order <i>beyond</i> arXiv:2607.18373 but extract only the <i>Hamiltonian</i> , not Fock-space observables                                                                                                          |
| Lublinsky & Mulian, arXiv:1610.03453, <i>High Energy QCD at NLO: from light-cone wave function to JIMWLK evolution</i> — this is Ref. [19] of the paper                                                                                                                                                 | Soft LCWF to third order in $g$ ; Fock space of one-gluon, two-gluon and $qq^-$ states                           | (!) <b>READ THIS FIRST.</b> The paper's own novelty claim is that [19] "did not contain all contributions at order $g^2$ ". That claim is the load-bearing one; see <a href="#">10-j-hep-assessment.md §3.1</a>                                                                           |
| Kovner et al., arXiv:2412.05085, <i>Born–Oppenheimer RG for High Energy Scattering: the Setup and the Wave Function</i>                                                                                                                                                                                 | Same Born–Oppenheimer framing as arXiv:2607.18373                                                                | Nearest methodological neighbour; check how much of the $0(g^2)$ operator is already there                                                                                                                                                                                                |
| Kovner & Lublinsky, <a href="#">hep-ph/0609227</a> <i>One gluon, two gluon: multigluon production via high energy evolution</i> (Ref. [63] of the paper); <a href="#">hep-ph/0608258</a> <i>Treading on the cut</i> ; arXiv:0901.2560 <i>Inclusive Gluon Production in the QCD Reggeon Field Theory</i> | Multi-gluon <b>production</b> from the high-energy evolution operator, including Pomeron loops                   | (!) <b>READ THIS FIRST.</b> This is the closest existing work to "moments of the multiplicity operator from an evolution operator". It is at LO/RFT level rather than NLO-LCWF level, but a referee will absolutely ask how your work differs. You must be able to answer in one sentence |
| Armesto, Dominguez, Kovner, Lublinsky, Skokov, JHEP 05 (2019) 025025, <i>The CGC density matrix: Lindblad evolution, entanglement entropy and Wigner functional</i>                                                                                                                                     | Reduced density matrix of the soft sector; entanglement entropy; Wigner functional                               | <b>Highly relevant.</b> They already treat the soft gluon sector as a quantum state with a density matrix. Multiplicity moments are a natural observable of that density matrix — check whether they computed any                                                                         |
| arXiv:2002.02282, <i>JIMWLK Evolution, Lindblad Equation and Quantum–Classical Correspondence</i>                                                                                                                                                                                                       | Decoherence framing of JIMWLK                                                                                    | Supports the "quantum vs classical fluctuation" decomposition of §1.8                                                                                                                                                                                                                     |
| Kovner, Lublinsky, Serino, <i>Entanglement entropy, entropy production and time evolution in high energy QCD</i> , PLB 792 (2019) — Ref. [61] of the paper                                                                                                                                              | Coherent-operator diagonalization at $0(g)$                                                                      | The paper's §6 extends this by adding the $0(g)$ three-gluon operator $C$                                                                                                                                                                                                                 |

**Assessment of this block.** The  $0(g^2)/0(g^3)$  wave function is *not* virgin territory. The gap is specific and real: **nobody has taken these wave functions and computed the factorial cumulants of the gluon number operator**. The formalism exists; the observable does not.

### 7.2 Multiplicity distributions in the CGC/Glasma — the classical result

| Work                                                                                              | Content                                                                                                                      | Status                                                                                |
|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Gelis, Lappi, McLerran, <i>Glittering Glasma</i> , Nucl. Phys. A 828 (2009)                       | $P(N)$ from Glasma flux tubes = <b>negative binomial</b> , with $k \propto (N_{c^2}-1) Q_{s^2} S_{\perp} / 2\pi$             | <b>The established result.</b> Classical order. Your $c_2$ at LO must reproduce $1/k$ |
| Lappi and collaborators, various                                                                  | NBD parameter from CGC, comparison to RHIC/LHC                                                                               | Established                                                                           |
| arXiv:1209.4105, <i>KNO scaling from a nearly Gaussian action for small-<math>x</math> gluons</i> | KNO emerges if the effective action for colour charges at $\sim Q_s$ is nearly Gaussian; needs saturation + running coupling | Directly relevant to Part IV §4.6                                                     |

| Work                       | Content                                                                     | Status                                                         |
|----------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------|
| IP-Jazma, arXiv:1808.01276 | Critical assessment of saturation explanations of small-system collectivity | Useful for calibrating claims about phenomenological relevance |

**Assessment.** The classical/LO multiplicity distribution of the CGC is **thoroughly done**. Any paper that produces an NBD from a CGC calculation and stops there is not publishable. The novelty must be the  $0(g^4)$  correction, the sign structure, or the angular discriminator.

### 7.3 Dipole models, KNO, and entanglement

| Work                                                                                        | Content                                                                                                 |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Mueller (dipole cascade); Le, Mueller, Munier, PRD 104 (2021) 056025                        | Multiplicity distribution of dipoles from the Le-Mueller-Munier equation; generating-function formalism |
| Kharzeev & Levin, PRD 95 (2017) 114008; arXiv:2110.04881                                    | DIS entanglement entropy $S = -\ln N(x, Q^2)$ ; maximally entangled proton; $P_n$ geometric             |
| arXiv:2302.01380, <i>Universality of KNO scaling in QCD at high energy and entanglement</i> | KNO from entanglement; comparison to H1 and ALEPH                                                       |
| arXiv:2406.04856                                                                            | KNO scaling emergence in LHC jets                                                                       |

**Assessment.** Generating functions and multiplicity distributions in small- $x$  QCD are a *busy* field, but almost entirely in the **dipole-cascade / partonic-counting** language, where the "multiplicity" is a number of dipoles evolving by a Markov branching process. That is a fundamentally different object from the Fock-space gluon number of a light-cone wave function. **This is a genuine gap and a genuine opportunity**, but it is also a hazard: a referee from that community will ask how your  $P(N)$  relates to the Mueller/Kharzeev-Levin  $P_n$ . Have an answer. (Mine: theirs is a classical branching process with geometric statistics; yours is a Gaussian quantum state with Poisson statistics at LO — they are *different* distributions and the difference is precisely the coherence of the emitted field. Saying this clearly is itself a contribution.)

### 7.4 Correlations, Bose enhancement, the ridge

| Work                                                                                                 | Content                                                                                                                                                | Overlap                                                                               |
|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Altinoluk, Armesto, Beuf, Kovner, Lublinsky, arXiv:1503.07126, <i>Bose enhancement and the ridge</i> | Bose enhancement of gluons in the projectile WF $\rightarrow$ azimuthal collimation; correlations suppressed by $1/(S_\perp \times N_{\text{gluons}})$ | <b>High.</b> This is the $\alpha^* n \alpha$ term of §4.2. (!) <b>READ THIS FIRST</b> |
| arXiv:1804.02910, <i>Correlations and the ridge in the CGC beyond the glasma graph approximation</i> | Beyond glasma graphs                                                                                                                                   | High                                                                                  |
| arXiv:1612.07790, <i>Exploring correlations in the CGC wave function: odd azimuthal anisotropy</i>   | Odd harmonics from the CGC wave function                                                                                                               | Relevant to the odderon/( $ppp$ ) project                                             |
| arXiv:1808.04982, <i>Energy evolution and Bose-Einstein enhancement for double parton densities</i>  | Evolution of the Bose-enhanced double-parton density                                                                                                   | Relevant to $Y$ -dependence of $c_2$                                                  |

**Assessment.** The **near-side** ( $k \approx +q$ ) correlation in the projectile wave function is well-established and thoroughly explored. **The away-side** ( $k \approx -q$ ) correlation from the pair-emission/squeezing kernel  $B_2$  of the NLO wave function is, as far as I can determine, **not computed anywhere**. That is the sharpest open item in the survey.

Caveat to that claim: away-side correlations from momentum conservation appear in many guises in this literature, and it is possible that the  $\alpha^* \alpha^* m$  structure appears implicitly in a glasma-graph calculation without being identified as a squeezing effect. **Verify by reading 1503.07126 and 1804.02910 carefully before claiming novelty.**

### 7.5 Coherent states, squeezed states, quantum optics

| Work                                                                                                      | Content                                            | Overlap                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| arXiv:2605.03038, <i>Squeezed-state radiation in shockwave scattering: QCD-Gravity double copy</i> (2026) | Squeezed-state radiation from shockwave scattering | <b>(!) HIGHEST-RISK ITEM IN THE SURVEY.</b> Recent, directly on the squeezed-state-in-QCD theme. Read before doing anything else |

| Work                                                                                                      | Content                                                                               | Overlap                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">hep-ph/9604371</a> , <i>Squeezed States and Particle Production in High Energy Collisions</i> | Squeezed-state phenomenology of multiplicity distributions                            | Old, phenomenological, not CGC. Low overlap but must be cited                                                                                                            |
| <a href="#">hep-ph/9902402</a> , <i>Gluon Squeezed States in QCD Jet</i>                                  | Squeezed gluon states in jets                                                         | Low overlap                                                                                                                                                              |
| Quantum-optics standard results                                                                           | Displaced squeezed thermal states, photon-number distributions, Fredholm determinants | <b>This is where §2.2 comes from.</b> Cite it properly — the generating function is a known quantum-optics result and presenting it as new would be an easy referee kill |

**Assessment.** The coherent-state picture of the classical CGC field is folklore and is stated in many places (including the paper's own Eq. (3.8)). The **squeezed**-state picture — that  $\Omega|0\rangle$  is a *displaced squeezed* state and that this is what generates non-Poissonian statistics — is, in the CGC context, either new or at most touched on in arXiv:2605.03038. **This framing is the most marketable aspect of the project**, precisely because it makes the result legible to people outside small- $x$  QCD. It is also the aspect most vulnerable to being scooped.

## 7.6 Summary: known vs open

### Known (do not claim as new):

1. LO CGC soft gluon state = coherent state;  $dN/d^2k = WW$  field.
2.  $P(N)$  from classical CGC/Glasma = negative binomial;  $k \propto (N_{c^2}-1)Q_{s^2}S_{\perp}$ .
3. Near-side Bose enhancement and HBT in the projectile wave function  $\rightarrow$  the ridge.
4. NLO JIMWLK Hamiltonian and the  $0(g^3)$  soft wave function (Kovner–Lublinsky–Mulian).
5. Multiplicity generating functions and KNO in the dipole-cascade language.
6. Entanglement-entropy-based  $P_n$  (Kharzeev–Levin) and its data comparisons.
7. Displaced-squeezed-state photon statistics (quantum optics; import, do not rederive).
8. CGC reduced density matrix, Lindblad/decoherence framing of JIMWLK.

### Open (candidate contributions):

1. **Factorial cumulants of the Fock-space gluon number operator from an NLO CGC evolution operator.** Not done anywhere I can find. The formalism exists; the observable does not.
2. **The  $0(g^4)$  quantum correction to the negative-binomial parameter  $k$ .** Not done.
3. **The identification of  $\Omega|0\rangle$  as a displaced *squeezed* state, and of the displacement–squeezing interference as the leading source of non-Poissonian statistics.** Possibly new; check arXiv:2605.03038.
4. **The away-side ( $k \approx -q$ ) two-gluon correlation from  $B_2$ ,** and its contrast with near-side Bose enhancement. Sharpest open item; verify against 1503.07126.
5. **A possible sub-Poissonian ( $C_2 < 0$ ) region** — non-classical gluon statistics in QCD. Would be genuinely striking; nobody has looked.
6. **The  $0(g^3)$  odderon contribution ( $ppp$ ) to the gluon multiplicity.** Small, clean, unexplored.
7. **KNO scaling and its  $\alpha_s$ -suppressed violation from the NLO wave function** (as opposed to from the dipole cascade). Open, but crowded neighbourhood.

### Closed by prior art, do not pursue:

- Reproducing the NBD from CGC.
- Near-side ridge from Bose enhancement.
- Three-gluon correlations (both incomplete here *and* partly covered by Kovner–Lublinsky multi-gluon production).

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**Sources:** [arXiv:2607.18373](#) · [arXiv:1610.03453](#) · [arXiv:2412.05085](#) · [hep-ph/0609227](#) · [hep-ph/0608258](#) · [arXiv:0901.2560](#) · [JHEP 05 \(2019\) 025025](#) · [arXiv:2002.02282](#) · [Glittering Glasma](#) · [arXiv:1209.4105](#) · [arXiv:1808.01276](#) · [PRD 104 \(2021\) 056025](#) · [PRD 95 \(2017\) 114008](#) · [arXiv:2110.04881](#) · [arXiv:2302.01380](#) · [arXiv:2406.04856](#) · [arXiv:1503.07126](#) · [arXiv:1804.02910](#) · [arXiv:1612.07790](#) · [arXiv:1808.04982](#) · [arXiv:2605.03038](#) · [hep-ph/9604371](#) · [hep-ph/9902402](#)



# Part VIII — Publication potential: a critical assessment

You asked for honesty rather than encouragement. Here is my actual read.

## 8.0 The blunt summary

There is one good paper here, not three. The good paper is:

*Non-Poissonian gluon statistics from the NLO CGC wave function: factorial cumulants, the negative-binomial parameter, and away-side pair correlations.*

It is a **PRD paper**. It is not a JHEP paper as currently scoped, and it is more than a short note. Getting it to JHEP requires adding produced-particle observables (target Wilson lines), and that roughly doubles the work.

The three things that make it publishable rather than routine:

- 1.  $C_2$  is **complete at  $O(g^4)$**  with the operator that already exists — a finished, checkable statement, not a partial result.
- 2. The **displaced-squeezed-state identification** makes the result legible and quotable outside the small- $x$  community.
- 3. The **away-side vs near-side discriminator** gives it a connection to the ridge literature, which is where the citations are.

The three things that could sink it:

- 1. The new effect may be numerically negligible against classical colour-charge fluctuations (test P3 in Part VI — run it first).
- 2. Gluon number in a light-cone wave function is not an observable (Part III §3.5).
- 3. arXiv:2605.03038 may already contain the squeezed-state framing.

## 8.1 Ranked projects

### Project A — $O(g^4)$ factorial cumulants and the NBD parameter ★★★★★

*Compute  $\langle N \rangle$  to  $O(g^4)$  and  $C_2$  complete at  $O(g^4)$ ; evaluate in MV; extract  $k_{\text{NBD}}$  and its quantum correction; demonstrate JIMWLK consistency of the  $Y$ -dependence.*

**Novelty.** Real but incremental. The wave function exists (yours); the observable is new; the quantum-optics machinery is imported. Honest characterization: *first computation of a Fock-space fluctuation observable from an NLO CGC evolution operator*. That is a legitimate "first", and the completeness of  $C_2$  at  $O(g^4)$  is what makes it a paper rather than a note.

**Technical difficulty.** Moderate. The Gaussian reduction (Part I §1.5) makes the Fock algebra trivial — and it needs only Eqs. (5.1)–(5.3) of the paper. The real work is (i) the  $p$ -ordering bookkeeping, (ii) the colour algebra of  $\langle p^4 \rangle$ , (iii) the IR analysis. **3–5 months of focused work** for someone who already owns the  $\Omega$  derivation.

**Expected referee objections**, with the fix:

| Objection                                                            | Severity | Fix                                                                                                                      |
|----------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------|
| "Gluon number in the LCWF is not an observable"                      | High     | Show the $Y$ - and scheme-dependence is JIMWLK; frame as a wave-function property analogous to the ugd. Mandatory        |
| "Your effect is swamped by classical colour fluctuations"            | High     | Part V Model 7: quantify the ratio explicitly; use the angular discriminator                                             |
| "The generating function is textbook quantum optics"                 | Medium   | Cite it as such; the contribution is the QCD kernels, not the Gaussian algebra. Do not oversell                          |
| "How does this differ from Kovner–Lublinsky multi-gluon production?" | Medium   | Prepare a one-paragraph answer. Theirs: production cross sections at LO/RFT. Yours: wave-function Fock statistics at NLO |
| " $C_2$ is IR-divergent / cutoff-dependent"                          | Medium   | Quote $k_{\perp}$ -differentially; show which combinations are finite                                                    |
| "The $p$ -commutator terms are dropped/mishandled"                   | Medium   | Appendix demonstrating ordering-independence                                                                             |

| Objection                                            | Severity | Fix                                                                                                                      |
|------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------|
| "You use only three of the six coefficient families" | Medium   | True, and unavoidable: $C, D, E$ annihilate $ \theta\rangle$ . Say so, and point to Project E as their consumer          |
| "Why not $C_3, C_4$ ?"                               | Low      | Because the $O(g^3)$ operator is needed. Say so in the text — pre-empting this converts a weakness into evidence of care |

**Additional calculations required.** The  $p$ -ordering appendix; the JIMWLK consistency check; the IR analysis. All are within the existing formalism.

**Is numerical work sufficient?** No. A purely numerical paper here would be rejected or heavily criticized — the community expects the analytic  $O(g^4)$  expression. **You need the closed-form  $C_2$  in terms of  $A_2$  and  $B_2$  with the colour algebra done, plus MV numerics.** Numerics alone is a note; analytics alone is publishable but thin. Both together is a PRD.

**Target: PRD. Realistic. Verdict: do this one.**

### Project B — Extend $\Omega$ to $O(g^3)$ and compute $C_3, C_4$ completely ★★★★★

**Novelty.** High if completed — nobody has the complete third factorial cumulant.

**Technical difficulty. Severe.** You must compute the three-gluon component of  $\Omega|\theta\rangle$  at  $O(g^3)$ , which is the genuine two-loop-level LCPT calculation that Kovner–Lublinsky–Mullian did for the Hamiltonian. Reproducing it in normal-ordered Fock form is a substantial project on its own. **12–24 months.**

**Expected objections.** "You have redone KLM in a different basis." That objection is fatal unless you can show the Fock-space form contains information the Hamiltonian does not — which it does (the Hamiltonian is the diagonal matrix element; the wave function has off-diagonal structure), but you must argue it explicitly.

**Verdict: this is a *second* paper, and only after Project A lands.** Do not attempt it first. The risk of spending a year and being told "this is KLM" is real.

### Project C — Away-side two-gluon correlation from the squeezing kernel ★★★★★

Compute  $\Gamma(k, q)$  at  $O(g^4)$ , decompose into near-side (Bose) and away-side (pair-emission) components, predict the  $\Delta\phi$  structure.

**Novelty.** Potentially the highest in the whole program — **if** it is not already implicit in the glasma-graph literature. Verify against arXiv:1503.07126 and arXiv:1804.02910 first.

**Technical difficulty.** Low-to-moderate *given Project A* — it is the same kernels, differently projected. Essentially free once  $A_2$  and  $B_2$  are in hand.

**Expected objections.** "Away-side correlations from momentum conservation are trivial/background." This is the serious one. The defence must be quantitative: the *normalization*, the  $Q_s$ -scaling, and the fact that this particular away-side term is  $O(g^4)$  while the near-side Bose term is  $O(g^6)$  — i.e. at NLO the away side should *dominate*, which is not what a trivial momentum-conservation argument would give.

**Verdict: fold this into Project A as its second half.** Do not publish separately — alone it is thin, but as the physics payoff of Project A it is what makes the paper interesting. **This is what turns Project A from ★★★ to ★★★★★.**

### Project D — $O(g^3)$ odderon contribution to the gluon multiplicity ★★★★★

Compute  $2\text{Re}f_{A_2^{\{1\}}A_2^{\{2\}}}$ , **requiring**  $\langle ppp \rangle \propto d^{\{abc\}}$ ; show it vanishes in MV and evaluate in a C-odd model.\*

**Novelty.** Genuinely unexplored, and it connects to the active odderon programme.

**Technical difficulty.** Low. This is a contained calculation — **2–3 months.**

**Expected objections.** "The odderon contribution to an unobservable quantity is doubly unobservable." Fair and hard to answer. Also: "the C-odd correlator is model-dependent."



**Verdict: a good letter or a section of Project A, not a standalone paper.** As a standalone it is a short note in PRD or EPJC at best. As a section of Project A it adds genuine value at low cost. **Recommend folding in.**

### Project E — Produced-gluon cumulants (with target Wilson lines) ★★★★★ (impact) / ★★★★★ (feasibility)

*Scatter the NLO dressed state off a CGC target; compute cumulants of the produced gluon multiplicity.*

**Novelty.** High, and it solves the observability problem outright. **It is also the natural consumer of the paper's  $C_2$ ,  $C_3$ ,  $C_4$  coefficients**, which play no role in  $\Omega(\theta)$  but contribute the moment  $a \rightarrow U a$ .

**Technical difficulty. High.** You need  $\langle U U U U \rangle$  (quadrupole) and higher target correlators for  $C_2$ ,  $\langle U^6 \rangle$  for  $C_3$ . The Wilson-line correlators are known in the Gaussian/MV approximation but the bookkeeping is heavy, and the NLO real/virtual cancellation in the *production* amplitude is a known hard problem (the NLO  $pA$  production literature is littered with negativity issues and scheme subtleties).

**Expected objections.** All the standard NLO-production objections — rapidity subtraction scheme, negative cross sections, kinematic constraints. This is a mature and contentious subfield.

**Verdict: this is the JHEP version, and it is a 2-year project.** High reward, high risk. **Do not start here.** Reassess after Project A. Note the paper's own conclusion already announces single inclusive gluon production at NLO as work in progress (Ref. [53]), so coordinate.

### Project F — KNO scaling and its violation from the NLO wave function ★★★★★

**Novelty.** Moderate. Crowded field (dipole cascade, entanglement, nearly-Gaussian action).

**Difficulty.** Low given Project A — it is a replot.

**Objections.** "KNO in small- $x$  QCD has been derived several ways already; what does the wave function add?" Hard to answer convincingly, because the dipole-cascade derivations are closer to the measured quantity.

**Verdict: a figure and a paragraph in Project A. Not a paper.**

### Project G — Purely numerical study of $P(N)$ in MV ★★★★★

**Verdict: not publishable alone.** Without the analytic  $O(g^4)$  result this is a Monte Carlo exercise reproducing a known NBD. A referee will say exactly that. **Do not write this paper.**

## 8.2 Ranking

| Rank | Project                        | Target     | Time     | Risk   | Recommendation                        |
|------|--------------------------------|------------|----------|--------|---------------------------------------|
| 1    | A + C + D combined             | PRD        | 6–9 mo   | Medium | <b>Do this.</b> One substantial paper |
| 2    | E (produced gluons)            | JHEP / PRD | 18–24 mo | High   | Second paper, after A                 |
| 3    | B ( $O(g^3)$ operator, $C_3$ ) | JHEP / PRD | 12–24 mo | High   | Third paper, or never                 |
| 4    | F (KNO)                        | —          | —        | —      | Section of A                          |
| 5    | G (numerics only)              | —          | —        | —      | <b>Do not</b>                         |

## 8.3 The honest bottom line

**Project A+C+D is publishable in PRD with roughly 70% confidence**, conditional on the go/no-go test (Part VI, P3) coming out favourably. If the quantum contribution to  $c_2$  turns out to be  $< 1\%$  of the classical one across all realistic  $Q_s$  and  $k_{\perp}$ , then the paper becomes "we computed a correction that does not matter", which is publishable but low-impact, and I would downgrade to ★★☆☆☆ and suggest reframing entirely around the away-side angular structure (where the classical

contribution is absent, so a small absolute size does not matter).

**It is not a JHEP paper as scoped.** JHEP in this subfield expects either a substantially new formalism or an observable-level calculation. Project A is neither — it is a clean, complete, first-of-its-kind observable extracted from an existing formalism. That is PRD.

**It is more than a short note**, provided the analytic  $O(g^4)$  expression, the colour algebra, the IR analysis, the JIMWLK check, and MV numerics are all present. Drop any two of those and it becomes a note.

**The largest single risk is not technical — it is scooping.** arXiv:2412.05085 (Kovner et al., Born–Oppenheimer RG) and arXiv:2605.03038 (squeezed-state radiation) are both recent and both adjacent. The group that wrote the NLO JIMWLK papers has every tool needed to do Project A in a few months if they decide to. **Read those two papers this week, and if the ground is clear, move fast.**

# Part IX — Research roadmap

Three projects, in the order they should be attempted. Times assume one person working on this as their main project, with the  $\Omega$  derivation already in hand.

## Phase 0 — Two weeks. Decide whether to proceed at all.

This phase costs almost nothing and can kill the project before you invest in it. **Do not skip it.**

| #   | Task                                                                                                                                          | Output                                                                                                                                |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 0.1 | Read <a href="#">arXiv:2605.03038</a> (squeezed-state shockwave radiation) and <a href="#">arXiv:2412.05085</a> (Born–Oppenheimer RG) in full | Scooping assessment. If either contains the displaced-squeezed identification <i>and</i> multiplicity statistics, reframe immediately |
| 0.2 | Read <a href="#">arXiv:1503.07126</a> and <a href="#">arXiv:1804.02910</a> , looking specifically for an $(aa)$ -type (anomalous) correlator  | Determines whether the away-side result (Project C) is new                                                                            |
| 0.3 | Read <a href="#">Lublinsky–Mulian 1610.03453</a> (= Ref. [19]), looking for the two-gluon Fock amplitude                                      | Determines how much of $B_2$ is already public                                                                                        |
| 0.4 | Read <a href="#">hep-ph/0609227</a> ( <i>One gluon, two gluon</i> , = Ref. [63])                                                              | Prepare the one-paragraph "how is this different" answer                                                                              |
| 0.5 | Implement Model 2 (1-D toy) + the Fredholm-determinant generating function; verify against the four quantum-optics limits in §2.2             | Working, validated code. ~2 days                                                                                                      |
| 0.6 | Run <b>Model 5 (translationally invariant)</b> with the actual $B_2$ from Eq. (5.3); evaluate $C_2$ and compare to $\text{Var}_\rho[\mu]$     | <b>THE GO/NO-GO NUMBER.</b> If quantum/classical $< 10^{-2}$ everywhere, reframe around angular structure                             |

**Decision point.** Proceed to Phase 1 only if 0.1–0.4 leave the ground clear and 0.6 is not catastrophic.

## Project A(+C+D) — "Non-Poissonian gluon statistics from the NLO CGC wave function"

### 1. Required derivations

| #   | Derivation                                                                                                                                                                                                | Difficulty                                      | Time   |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|--------|
| A1  | Read $\alpha, m, n$ off $\Omega 0\rangle = N 0\rangle + \int A_2 a^\dagger 0\rangle + \int \int B_2 a^\dagger a^\dagger 0\rangle$ ; confirm $C, D, E$ drop out                                            | Easy                                            | 1 wk   |
| A2  | Verify unitarity relations ( $A_1 = -A_2^*, B_1 = -B_2^*, uu^\dagger - vv^\dagger = 1$ ) against Eqs. (4.4)–(4.10)                                                                                        | Easy — but it is your main internal cross-check | 1 wk   |
| A3  | $\rho$ -ordering: fix a prescription, compute the commutator-induced $O(g)$ reshuffles (cf. the paper's Eqs. 4.28–4.29), prove observable-level ordering independence for $\langle N \rangle$ at $O(g^4)$ | <b>Hard, error-prone</b>                        | 4–6 wk |
| A4  | $\langle N \rangle$ to $O(g^4)$ : assemble real ( $\int  B_2 ^2$ ), virtual ( $ \langle N \rangle ^2$ ), and interference terms                                                                           | Medium                                          | 3 wk   |
| A5  | <b>JIMWLK consistency:</b> show the $\ln(1/x)$ divergence of $A_4 = H_{\text{JIMWLK}} \triangleright \langle N \rangle_{L0}$                                                                              | Medium — high payoff                            | 3 wk   |
| A6  | $C_2 = 2\text{Re}\int A_2^* B_2 A_2^* + \int  B_2 ^2$ with full colour algebra; MV Wick reduction of $\langle \rho^4 \rangle$                                                                             | Medium                                          | 4 wk   |
| A7  | IR analysis: which of $\langle N \rangle, C_2, c_2, C_2(k_\perp)$ are IR-finite                                                                                                                           | Medium — determines the claims                  | 2 wk   |
| A8  | $\Gamma(k, q)$ decomposition into near-side ( $\alpha^* \alpha$ ) and away-side ( $\alpha^* \alpha^* m$ )                                                                                                 | Easy given A6                                   | 2 wk   |
| A9  | $O(g^3)$ odderon term (Project D): $2\text{Re}\int A_2^* \{(1)^*\} A_2^* \{(2)\}$ with $\langle ppp \rangle \propto d^{\{abc\}}$                                                                          | Easy                                            | 3 wk   |
| A10 | Gaussian-state null prediction for $\Gamma_3$ (three-body factorization up to $O(g^6)$ )                                                                                                                  | Easy                                            | 1 wk   |

**Total derivation time: ~6 months**, dominated by A3.

### 2. Required numerics

| #  | Computation                                                   | Model                      | Cost            |
|----|---------------------------------------------------------------|----------------------------|-----------------|
| N1 | Fredholm-determinant $P(N)$ code, validated                   | 2, 1                       | done in Phase 0 |
| N2 | $c_2$ vs $Q_s$ , quantum/classical decomposition ( $P3, P2$ ) | 5, then 4b                 | CPU-hours       |
| N3 | $c_2$ vs $\Lambda_{IR}$ and vs $k_\perp$ ( $P5, P7$ )         | 5                          | CPU-hours       |
| N4 | $\Gamma(\Delta\phi)$ near/away decomposition ( $P8$ )         | 5, 4b                      | CPU-hours       |
| N5 | Sign map in $(k_\perp, Q_s)$ ( $P12$ )                        | 5                          | CPU-hours       |
| N6 | Full MV lattice sampling, $P(N)$ , KNO ( $P1, P10$ )          | 4b, $L=512, N_{conf}=10^5$ | ~1 GPU-day      |
| N7 | $N_s$ -scaling validation ( $P9$ ), large- $N_c$ check        | 3, 6                       | minutes         |
| N8 | $Q_s$ -fluctuation robustness ( $P7$ -model)                  | 7                          | minutes         |

**Total numerics time: ~2 months** including code development, run mostly in parallel with the derivations.

### 3. Estimated difficulty

**Moderate-to-hard.** No single step is beyond a competent practitioner; the risk is accumulated algebra error in A3/A6. Mitigate with the 1-D toy (Model 2) and the  $N_s$ -scaling check (N7) as independent validations, and with the JIMWLK check (A5) as the global validation.

### 4. Estimated completion time

**6–9 months** to a submittable draft. Add 2–3 months if A3 goes badly (it might).

### 5. Publication target

**Physical Review D**, regular article, ~30 pages with appendices. Fallback: EPJC. Stretch: JHEP only if Project E material is folded in — do not plan on it.

### 6. Risks

| Risk                                               | Probability | Impact | Mitigation                                                         |
|----------------------------------------------------|-------------|--------|--------------------------------------------------------------------|
| Scooped by the KLM group or the 2605.03038 authors | Medium      | Fatal  | Phase 0.1; move fast; post to arXiv early                          |
| Quantum effect numerically negligible              | Medium      | High   | Phase 0.6 go/no-go; reframe around angular structure               |
| $p$ -ordering algebra error                        | Medium-high | High   | A2 unitarity check; 1-D toy; ordering-independence proof           |
| $c_2$ IR-divergent, no clean observable            | Medium      | Medium | A7 early; go $k_\perp$ -differential                               |
| Away-side result turns out to be known             | Medium      | Medium | Phase 0.2                                                          |
| "Not an observable" referee report                 | <b>High</b> | Medium | A5 (JIMWLK check) + explicit framing; accept one round of revision |
| "Uses only 3 of 6 coefficient families"            | Medium      | Low    | State it; point at Project E                                       |

### 7. Expected scientific impact

**Modest but real.** Realistic citation expectation: 15–40 over five years. It will be read by the small- $x$  formalism community (~50 people) and, *if* the squeezed-state and away-side framing lands, by the correlations/ridge community (~200 people). The  $k_{NBD}$  at NLO result has the longest legs, because it is the one number a phenomenologist can pick up and use.

It will **not** be a high-impact paper. It is a solid, careful, first-of-its-kind calculation that establishes a formalism others will use. That is a good outcome for a follow-up to a formalism paper, and it is the correct thing to write next.

## Project E — "Produced-gluon multiplicity cumulants at NLO" (second paper)

1. **Derivations.** Insert target Wilson lines:  $a \rightarrow U_a$ ; compute  $\langle U U U U \rangle$  (quadrupole) in the Gaussian/MV approximation; handle the rapidity subtraction and the real-virtual cancellation in the production amplitude; assemble  $C_2^{\text{produced}}$ . This is where the paper's  $C, D, E$  coefficients finally do work. **Hard.**
2. **Numerics.** Quadrupole correlators on a lattice (standard technology, e.g. existing JIMWLK codes);  $C_2^{\text{produced}}$  vs  $k_{\perp}, Y$ , target  $Q_s$ .
3. **Difficulty.** High. This is the contentious part of the NLO CGC literature.
4. **Time.** 18–24 months.
5. **Target.** JHEP or PRD.
6. **Risks.** Negativity/scheme problems endemic to NLO  $pA$  production; the possibility that the result is dominated by known glasma-graph physics; a crowded and critical referee pool. Also: coordinate with Ref. [53] (single inclusive production at NLO, in preparation).
7. **Impact.** Substantially higher than Project A *if it works* — this would be a genuinely observable NLO fluctuation prediction. Also substantially more likely to stall.

## Project B — "Third factorial cumulant and the $O(g^3)$ evolution operator" (third paper, or never)

1. **Derivations.** Extend  $\Omega$  to  $O(g^3)$ : the three-gluon component of  $\Omega|0\rangle$ , plus  $O(g^3)$  corrections to  $A_2$  and  $B_2$ . This is LCPT at the order Kovner–Lublinsky–Mulian worked at, but in normal-ordered Fock form. Then  $C_3$  complete at  $O(g^6)$ .
2. **Numerics.**  $C_3$  in the translationally invariant model (6-D) and MV (lattice).
3. **Difficulty.** High — the operator extraction is the whole job.
4. **Time.** 12–24 months.
5. **Target.** JHEP (the operator extension is formalism, which JHEP takes) or PRD.
6. **Risks.** "This is KLM in a different basis" is a real and possibly fatal objection. Also: by the time this is done,  $C_3$  may be of little interest if  $C_2$  turned out small.
7. **Impact.** Formal. Would be cited by the same ~50 people.

## Recommended sequence and honest expectation

```

Phase 0 (2 weeks) → Project A+C+D → PRD, ~9 months from now
                    ↓
                reassess: was C2 big? was the away-side result new?
                    ↓
yes → Project E (JHEP, +2 years)
no  → Project B (formalism, PRD/JHEP, +18 months)
        or move to a different problem

```

**One paper is near-certain. Two is plausible. Three is optimistic and I would not plan on it.**

The most valuable thing you can do in the next two weeks is Phase 0.6 — the go/no-go number. Every strategic decision in this document hinges on whether the  $O(g^4)$  quantum fluctuation is a percent-level or a per-mille-level effect, and that is currently unknown.

# Addendum — Will arXiv:2607.18373 make it into JHEP?

**This replaces an earlier version written before the paper was available.** That version estimated 45–55% and rested on three objections. Having now read v2 (57 pp., 28 Jul 2026), two of those objections are wrong and one is weaker than I stated. The estimate goes **up**.

The paper is already formatted "Prepared for submission to JHEP", so the venue question is really "will it survive refereeing there", not "is JHEP the right target". It is.

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## The short answer

**~65–75%, most likely after one round of moderate revision.**

The upgrade over the blind estimate comes from four things I could not see before:

1. **The novelty claim against the prior literature is specific and correct.** The paper does not claim to compute the wave function for the first time. It claims that Ref. [19] (Lublinsky–Mulian) obtained *some*  $O(g^2)$  and  $O(g^3)$  contributions but "did not contain all contributions at order  $g^2$ ", and that the present coefficients "complete and extend" them. That is a well-defined, checkable claim, and it is the right one. My earlier objection — "the wave function is already known to higher order, so why do  $O(g^2)$ ?" — **was wrong**: partial knowledge at a higher order does not imply complete knowledge at a lower one.
2. **The  $O(g)$  trilinear coefficient  $C$  makes the diagonalization a real result, not a check.** I assumed §6 would reproduce the standard coherent-state statement. It does not. The paper shows that the coherent operator used in Ref. [61] is *insufficient* at order  $g$ : the three-gluon vertex generates an operator  $C$  (structure  $a^\dagger a^\dagger a$ , order  $g$ , carrying no  $p$ ) at the same order, and only when  $C$  is included do the off-diagonal elements cancel identically, giving  $\Omega^\dagger H_1 \Omega = H_0$  exactly (Eq. 6.5). At  $O(g^2)$  the  $C \cdot C$  contributions cancel (Eq. 6.8) and the sole survivor is the  $p^2$  coherent background-field energy (Eq. 6.10). That is a correction to received wisdom, cleanly stated. It is still not a physical observable — see below — but it is more than internal validation.
3. **The volume and explicitness of the results.** Six coefficient families, given in closed form, with the  $D$  and  $E$  families expressed through  $A$  and  $C$  where possible, plus  $\Omega = \exp(iG)$  with  $G$  Hermitian by construction (Eqs. 5.11–5.12). JHEP rewards complete, reusable formula sets; this is one.
4. **Reception context.** DOE-funded, NC State, acknowledging Lublinsky and Skokov — i.e. the people whose line of work this extends — and a named independent cross-check of critical parts. None of that is a scientific argument, but it does mean the work has been seen by the community that supplies the referees, and that materially lowers the chance of a hostile report.

**What holds it below ~80%** is one surviving objection and three fixable presentation problems.

---

## The one objection that survives

**No physical observable is computed.** The paper's own conclusion is candid about this: the construction "constitutes a necessary step towards the computation of single inclusive gluon production at NLO", with double and triple gluon production and the multiplicity moments left to future work. So the "so what" is entirely deferred.

For JHEP this is survivable but not free. The standard form of the report is: *"a careful and useful technical construction, but the physics payoff is postponed; I would like to see at least one worked consequence."* Two ways to answer it, in order of cost:

- **Cheap (recommended).** Sharpen the argument for *why the normal-ordered form matters*, which the introduction currently gestures at rather than makes. The point is precise and strong: the existing NLO literature extracts the **Hamiltonian**, i.e. the diagonal matrix element of the evolution operator. That suffices for evolution equations and inclusive cross sections. It does **not** suffice for anything requiring off-diagonal Fock structure — multiplicity distributions, factorial cumulants, the reduced density matrix, entanglement measures. A normal-ordered  $\Omega$  gives those; a Hamiltonian cannot. One paragraph, and it converts "no result" into "a tool that unlocks a class of results".

- **Expensive.** Add a worked consequence. I earlier suggested holding the paper to add the  $O(g^4)$  variance. **I retract that.** At 57 pages with a clearly staged programme already announced, splitting is correct; adding two more pages of a different calculation would dilute the paper without changing the referee's basic reading.

## Three fixable problems that cost probability

### (a) The delineation against Ref. [19] is asserted, not shown — the highest-value fix

The claim "[19] did not contain all contributions at order  $g^2$ " appears once in the introduction and once in the conclusion. It is load-bearing for the entire novelty case, and the referee is asked to take it on trust. A referee who is an author of [19], or who has read it recently, will want specifics.

**Fix: a short table.** One column per coefficient family ( $N$ ,  $A_{12}$ ,  $B_{123}$ ,  $C_{12}$ ,  $D_{123}$ ,  $E_{123}$ ), one column saying "new / present in [19] / partially in [19]". Half a page. In my judgement this single addition is worth more than anything else on this list — it converts the central claim from an assertion into something a referee can verify in five minutes.

### (b) Cutoff and scheme dependence is raised and then dropped

§2.3 states correctly that "physical observables should ultimately be independent of the arbitrary separation scale  $v$ ". But the coefficients themselves are explicitly cutoff-dependent — Eq. (5.4) carries  $\theta(p^+ - k^+ - \Lambda)$  and  $\theta(k^+ - \Lambda - p^+)$ , and  $N$  in Eq. (4.80) is  $\propto \log(v/\Lambda)$  — and nothing demonstrates or even sketches the  $v$ -independence. This will be asked. It does not need a full proof; a paragraph identifying where the  $v$ -dependence sits and how it is expected to cancel against the valence sector would defuse it. Leaving it silent invites a referee to treat it as an unexamined gap.

### (c) Presentation defects in a paper whose value is algebraic accuracy

See `../review/draft-review-omega-g2.md` for the full itemized list. The ones that matter most for refereeing:

- **Eq. (6.4) has a sign error** — the two  $A$ -terms are printed with opposite signs, making the bracket anti-Hermitian, so it cannot cancel the Hermitian  $H_g$  and Eq. (6.5) does not follow. This is in the section that carries the paper's application.
- **Eq. (4.55)** is missing a factor  $i$  and a factor  $1/\sqrt{(k^+p^+)}$ ; the error propagates into **Eq. (5.4)**, where two terms inside one bracket then differ dimensionally.
- **Eq. (4.36)** omits the instantaneous contribution that Eq. (4.35) promises and Eq. (5.3) actually uses.
- **Eq. (2.23)** has unbalanced brackets and one operator product in upright roman.
- The step **(6.6) → (6.7) → (6.8)** is asserted with no intermediate algebra, and **(6.8)** — that the  $C^\dagger C$  structure vanishes identically — is stated without demonstration. This is the largest referee-facing gap in the paper.
- **Consolidate the internal checks.** The paper contains genuinely strong validation: unitarity imposed order by order, matching across four independent Fock sectors (vacuum, one-, two-, three-gluon), and the two diagonalization results. But these are scattered across 40 pages. A short "consistency checks" subsection listing them together would materially change how a referee experiences 57 pages of algebra — from "I cannot verify this" to "the author has verified this in four independent ways."

## What is *not* a problem

- **Length.** JHEP has no limit and is the right home for a long coefficient derivation.
- $N_f = 0$ . Standard for a first paper in a programme, and explicitly justified (gluonic dynamics dominate at small  $x$ ). Retract as a concern.
- **The "NLO" terminology.** I previously flagged a mismatch between "NLO" in the title and  $O(g^2)$ . **Retracted.** The abstract says "up to  $O(g^2)$ " in its second line, and the usage matches the established convention in this specific line of work — cf. the title of Ref. [19], *High Energy QCD at NLO: from light-cone wave function to JIMWLK evolution*. No referee in this community will be misled.
- **Single author.** Irrelevant.

## Recommendation

| Action                                                                     | Effect on acceptance                               | Cost       |
|----------------------------------------------------------------------------|----------------------------------------------------|------------|
| Fix Eq. (6.4), (4.55)/(5.4), (4.36), (2.23)                                | <b>prerequisite</b> — these are errors, not polish | days       |
| Add the coefficient-by-coefficient comparison table vs Ref. [19]           | <b>largest single gain</b>                         | half a day |
| Add a "consistency checks" subsection consolidating the four validations   | high                                               | half a day |
| One paragraph: why the normal-ordered form gives what a Hamiltonian cannot | high                                               | an hour    |
| Show the (6.6) $\rightarrow$ (6.8) algebra in an appendix                  | high                                               | 1–2 weeks  |
| A paragraph on $\Lambda/v$ -dependence and expected cancellation           | medium                                             | a day      |
| Hold the paper to add the $O(g^4)$ variance                                | <b>negative</b> — do not                           | months     |

With the errors fixed and the first three additions done, I would put it at ~80%. As it stands, ~65–75%.

**The most likely bad outcome is not rejection — it is a referee who asks for the Ref. [19] delineation and the scheme-dependence discussion, and the paper spends four months in revision that a day of work now would avoid.**

## One thing worth knowing for the follow-up

Reading the operator changes the plan in Part IX in one respect.  $\Omega|0\rangle$  is exactly  $N|0\rangle + \int A_2 a^\dagger|0\rangle + \int \int B_2 a^\dagger a^\dagger|0\rangle$  through  $O(g^2)$  — the **C, D, E** families annihilate the soft vacuum. So the multiplicity-moments paper needs **only Eqs. (5.1)–(5.3)**, which is very good news for its feasibility, and slightly awkward news for its framing: a referee may observe that it uses three of the six coefficient families.

The natural answer is Project E: after eikonal scattering  $a \rightarrow U a$ , the annihilation operators no longer kill the state and **C, D, E** all contribute. **The produced-gluon observables are what the rest of this paper is for**, and saying so in the concluding remarks would strengthen both papers at once.